PERCI v2: A Scrutinised Re-analysis of the Uranus-Moon Penetrator Secondary Payload

Independent audit, probabilistic delivery–impact–thermal–power–link models, a novel-methods study, mission-level analysis against flown practice, and a proposed v2 design

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Abstract

The Penetrator for Environmental, Regolith, and Crater Investigation (PERCI) was proposed in a 2025 concept study as a ∼119 kg penetrator released from the Uranus Orbiter and Probe (UOP) flagship 4.4 h before closest approach to Umbriel. In that design the penetrator is retargeted by four STAR 5D solid motors (a 450 m/s requirement), impacts at 0.9 km/s and survives 30 min on a 140 W heater. In this paper every equation, table and budget of the study was audited, which produced 61 findings (4 critical, 21 high). We then rebuilt the delivery, penetration, thermal, power and link models as deterministic, tested, probabilistic codes and ran a novel-methods study. The report's Umbriel trajectory row is reproduced within 4% in impact speed, but the minimum-ΔV central hit needs only 323–328 m/s and arrives at 0.75–0.76 km/s. The 140 W heater was found to be an artefact of a battery-only thermal mass and a 1 W/K mount assumption: with an insulated core of ordinary heat capacity, 30 min needs 0 W and 2 h needs $\leq$24 W in 90% of sampled cases. The link as written does not close, because the flagship carries no S-band; at UHF the median 10 kbps margin is -7.5 dB at 5.4–6.5 Mm, and of the planned 20 GB of data only about 16 MB can be returned. No impact model is validated above 341 m/s in ice, and the report's 3.4–8.3 m depth is the stiff-ice corner of a distribution spanning 2–600 m. Finally, the wheel sizing for the 10°/s flip is 43–66× too small, but the flip itself is unnecessary because the minimum-ΔV burn is 67°, not 180°, from the impact velocity. Compared with flown and studied penetrators, PERCI is 2.2 times faster than the fastest full-scale ice test and 2.6 times heavier than any survivable penetrator built, and the class has no successful flight. The mission-level analysis adds that the penetrator is stored for 16.0 years before use, not 13.4; that holding its motors at an analogous storage temperature needs 1.4–13 W against a 2 W cruise allowance; that the SMAD small-satellite cost model gives a floor of $46–59 M, at the SIMPLEx cap; and that a 13 kg-class penetrator meets 2.5 of the 3.0 in-situ science rows for about one sixth of the launch-mass equivalent, at the price of a hard-ice median shock near 24,000 g. A v2 baseline is proposed with recommendations and a list of what remains unknown. The penetration physics at 0.75 km/s is at TRL 2–3 and is the dominant open risk.

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Nomenclature

Trajectory and Delivery Parameters	
ΔV	retarget velocity change [m/s]
v_{imp}	impact speed [km/s]
T	release time before closest approach [h]
$g = h/(R\,v)$	signed aim parameter (h impact parameter, R moon radius, v relative speed)
P(acceptable)	probability of an acceptable impact (criteria in Section 4)
Penetration Parameters	
F	resisting force on the penetrator [N]
z	penetration depth [m]
a, ψ	penetrator radius [m] and nose caliber-radius-head
N, f_{nose}	nose shape factor and nose factor
R_t, Y, E, ρ_t	target resistance, yield strength, Young's modulus [Pa] and density [kg/m^3]
m, α	penetrator mass [kg] and incidence angle
Thermal Parameters	
C, C_s	core and shell heat capacity [kJ/K]
G, G_{ins}	conductance to the ice and core-to-shell conductance [W/K]
P	heater power [W]
k, c	thermal conductivity [W/m/K] and specific heat of the target
Risk Parameters	
S, P, D	severity, probability and detectability scores
RPN	risk priority number

1 Introduction

The Uranus Orbiter and Probe (UOP) flagship [1] would tour the Uranian system and make several ballistic flybys of the major moons. These flybys only allow remote observations, so in-situ measurements of surface strength and thermal properties remain out of reach of the orbiter alone. The 2025 PERCI concept study [2] proposed to fill this gap with a penetrator released from the flagship during a flyby and delivered to the surface of Umbriel.

However, a concept of this kind rests on a chain of models (delivery, impact, thermal, power and link), and an error in any one of them changes what the mission can return. The headline numbers of the 2025 study had not been independently checked, and several of its budgets disagree with each other. Nor had the concept been compared with the penetrators, impactors and landers that have flown or been tested, or completed with the mission-level elements that SMAD expects of any concept [3]: an operations timeline, a data and relay architecture, a reliability argument, a cost class, a verification flow and planetary protection. The goal of this paper is to address this gap. This study audits the 2025 report, rebuilds each of its models as deterministic, tested code with stated uncertainty, tests a set of new delivery and design ideas against the same baseline, adds the mission-level analysis with a comparison to current practice, and proposes a v2 design.

The rest of the paper is structured as follows. Section 2 describes the approach, the inputs and the literature used. Section 3 presents the audit of the 2025 report, and Section 4 the revised requirements and concept of operations. Section 5 gives the results of the delivery, impact, thermal, power, link and science models, together with the novel-methods study. Section 6 adds the mission-level analysis: heritage, deployment from the flagship, the operations timeline, data and relay, reliability and cold soak, cost, planetary protection and a science-per-kilogram trade. Section 7 discusses the recomputed risk register, the v2 design, the verification plan, the sources of error and model limitations, and the objections a reviewer would raise. The paper concludes with an assessment of the concept, recommendations and the work that should follow in Section 8.

1.1 Summary of Findings

Verdict. The concept is not invalidated by this re-analysis, but its headline numbers are not defensible as published. Four findings are critical: the link budget (B-01, B-02), the reaction-wheel sizing for the SRB flip (B-28) and the thermal 140 W sizing (A-01). The study also has real strengths. Its Umbriel trajectory row is independently reproduced, its power-budget arithmetic is internally consistent, its STAR 5D data match the manufacturer's catalogue, and its risk-matrix arithmetic is consistent (the "OK" rows of Table 13).

Corrected facts. The corrections that change a reported value are as follows:

- **Impact.** The summary and the appendix give different impact numbers (6.5–16 MN and 2.7–3.5 m versus 5.4–10.4 MN and 3.4–8.3 m). The Tate–Alekseevski input R_t is 0.454 GPa by the report's own formula, not 0.6 GPa, and the Glen's-law "flow stress" is a creep law applied far outside its range (A-03, A-04).
- **Link.** The table's 3.3 bps "Shannon limit" comes from a unit slip (0.125 Hz used for 0.125 MHz); the correct figure is 842 kbps. The EIRP is 11.4 dBW, not 5.67 dBW, and the receive gain is 24.2 dBi, not 15.0 dBi. The claims of "15 dB at 10 kbps" and "1 dB at 2.14 Mbps" cannot both hold (B-02..B-05).
- **Power.** The 37 W average is not reproducible (we obtain 29.2 W in descent and 143.8 W post-impact), whereas the 705,145.5 J total is reproducible (29.175 W×4.25 h + 143.76 W×0.5 h) (B-13, B-14).
- **Mass.** The 159 kg figure is a *wet* mass compared against a *dry* 180 kg limit. With AIAA S-120 growth the wet mass is 191 kg (dry 163 kg) (B-33).
- ΔV**.** With the report's own masses the 4-motor capability is 483 m/s, not 476 m/s, and the margin disappears with 9% mass growth (B-22, B-23).

v2 design changes. The changes are detailed in Table 11. The ΔV requirement is lowered from 450 to 330 m/s, with a 22% capability margin at MEV mass, and the impact speed falls from 0.90 to 0.75 km/s. The 180° flip is removed and replaced by a slow 67° slew. The heater power drops from 140 W to 0 W for 30 min and to ≤24 W (90th percentile) for 2 h. The drag flare is replaced by a split forebody/aftbody with a surface antenna, and the link becomes a UHF 6 kbps link with a corrected budget. The science is descoped from interior structure and plume composition to near-surface strength, thermal conductivity and descent imagery.

Novel-methods workstream. In this workstream (Section 5.7) 14 ideas were simulated: 4 are kept, 5 are partial, 1 is unproven and 4 failed. The strongest numerical result is that release times of 12–24 h cut the retarget ΔV by a factor of 3–6 and the propulsion mass from 26 kg to 3–9 kg. However, P(acceptable impact) then depends on the unknown flyby orientation (0.47–1.00 at 24 h). We are strict about what is new: only the UOP/Umbriel-specific numbers and the robustness analysis. The flyby-released penetrator itself has prior art (Wurz et al., 27 h before closest approach).

Mission level and heritage. The mission-level analysis (Section 6) finds that no penetrator has yet returned data from another body, and that PERCI sits 2.2 times above the fastest full-scale ice test in speed and 2.6 times above the heaviest survivable penetrator in mass. The relay is a single 30-minute pass that returns 0.15 MB after impact at the median, while store-and-forward on the flagship costs only 1.8 h of Ka-band time. The motors need cruise heat that the v2 budget does not carry, the SMAD cost model puts the element at the SIMPLEx cap before any high-g qualification, and planetary protection needs an Umbriel-specific transport analysis or Category IV controls. A 13 kg-class penetrator is the strongest alternative found (Section 6.8).

Claims withdrawn or corrected in this revision. (i) The 13.4-year cold soak is replaced by 16.0 years of storage from launch to use. (ii) The SIMPLEx cost framing is not supported. (iii) The 0.64 W cruise load is not shown to be sufficient, because it has no motor heater. (iv) The 2047 encounter dates are placeholders, since the 2026 UOP update moves launch to 2035–2040 [4].

Remaining unknowns. The main unknowns (Section 7.4) are the penetration of 85 K ice/regolith at 0.75 km/s; the shock survival of a broadband seismometer and of the cells above 5,000 g; the UOP UHF antenna gain, G/T and tour state; the Umbriel regolith thickness; and the SRB behaviour after 16.0 years of storage.

2 Approach

The goal of this study is to test whether the 2025 PERCI concept closes once each of its models is recomputed independently, and to propose a design that does. The analysis links

1. the delivery trajectory from the flagship flyby to the moon,
2. the impact and penetration of the target,
3. the thermal, power and link budgets, which set how long the penetrator survives and how much data it returns, and
4. the science return that follows from these,

and every result is checked against the corresponding number in the 2025 report.

2.1 Inputs and Provenance

The inputs are the 85-page final report [2], the UOP study [1], and a verified literature and data survey kept in `research/`. The survey covers 53 sources, of which 38 were opened in full and 15 are snippet-only, and

each parameter in it is tagged VERIFIED, DERIVED or UNVERIFIED.

2.2 Method

Every number in this report is produced by deterministic code (`analysis/`, driven by a single `run_all.py` and checked by 61 unit tests), and every random draw is seeded. Where a parameter is not in the literature it is tagged ASSUMED in the code, and the result is reported as a band over the assumption, never as a point. Spreads are taken over three sources: the four unknown orientations of the UOP flyby geometry, Sobol-sampled target and material properties, and a fixed-seed Monte Carlo of the error budget.

2.3 Replicated and New Elements

Several parts of the analysis are replicated, standard methods: circular-restricted three-body shooting (Hill/Clohessy–Wiltshire dynamics [5]), cavity-expansion penetration (the Forrestal–Luk family [6, 7]), lumped-capacitance and radial conduction thermal models, link-budget bookkeeping and FMEA arithmetic. The elements that are new in this work are the quantified UOP/Umbriel delivery families and their robustness to the unknown flyby orientation (Section 5.1), the reverse-engineering of the report's thermal table (Section 5.3), the combined probabilistic acceptance model, and the model-referenced risk register. We make no claim of novelty beyond those and the audit itself.

2.4 Honesty Rules

No source or datum is invented in this paper. Items that could not be verified are labelled UNVERIFIED, for example the ice RF loss at 100 K and 2 GHz, the Forrestal–Luk constants for ice, whether UOP carries a spectrometer such as L'Ralph, the aerogel/MLI values, and the Uranian-moon ephemeris error in 2047.

2.5 Literature and Data

The key verified inputs are listed here; the full tables are in `research/params.json`. For Umbriel, $GM = 85.1 \pm 1.9$ km^3/s^2 and $R = 584.7$ km [8], the escape speed is 0.540 km/s, and the orbit has $a = 266{,}000$ km and $P = 4.144$ d [9]. For Uranus, $GM = 5.794 \times 10^6$ km^3/s^2 [10]. The UOP Umbriel flybys are on 22 Jun 2047 (5,354 km, 0.52 km/s) and 3 Oct 2047 (3,653 km, 0.73 km/s) [1]. The UOP study lists X-band, Ka-band and a single UHF low-gain antenna with a 6 kbps probe relay, and it includes no penetrator [1]. Umbriel's thermal inertia is 5–15 SI units, against 2000–2600 for solid ice [11], so the surface is porous regolith whose thickness is unknown [12].

Several assumptions in the report are contradicted by the literature (the full list is in `research/contradictions.md`):

1. An impact at 0.9 km/s into ice is 2.6× the fastest documented full-scale ice impact (20 kg at 341 m/s, 24,000 g [13]). Young's empirical equations are valid only below 1,219 m/s and carry no data in soft materials at very high speed [14]. Flown penetrators landed at 80–200 m/s (Mars 96, DS-2) [15, 16], and none has operated [17].
2. Crary waves need a floating ice shell above an ocean [18], and none is established for Umbriel.
3. The UOP orbiter has no S-band, and its UHF antenna is a probe-relay antenna [1].
4. An impact seismic efficiency of 10^{-4}–10^{-3} is optimistic for porous regolith: Apollo booster impacts into unbonded soil gave 10^{-6}–10^{-5}, and InSight adopted 5×10^{-4} [19].
5. For penetrator instruments, the Lunar-A seismometer survived 8,000–10,000 g [20], unprotected silicon MEMS suspensions fail at tens of g [21], and the DS-2 values are 30,000 g for the forebody and 80,000 g for the cells [16, 22].
6. The DS-2 and Mars 96 designs separate the forebody and aftbody so that the antenna remains at the surface [15, 16]; the report's single body with a drag flare has no heritage.

3 Audit of the 2025 Report

Every equation, table, budget, link budget, impact model, thermal result, ΔV, mass budget and timeline in the report was recomputed independently. The scripts are `audit/recompute_*.py`, and the paper-versus-recomputed output is kept in `audit/paper_vs_recomputed.txt`. Figure 1 gives the counts by severity, and the full list is given in Table 13 (Appendix A).

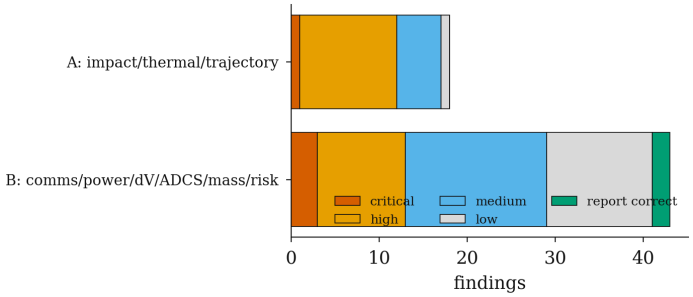


Figure 1: Audit findings by severity (61 total). Two of the findings (B-13 power sums, B-37 RPN arithmetic) confirm that the report is right.

Notes already known to the team, confirmed or refuted. Several issues were known to the original team before the audit; each was checked as follows.

1. The summary and appendix impact ranges disagree: *confirmed* (A-08).
2. The 37 W versus 29.2 W discrepancy: *confirmed*, and the 37 W has no reproducible origin (B-14).
3. T-4.6/T-4.4 h versus Table 2: *confirmed*; five different pre-impact times appear in the report (B-15, B-40).
4. The link table shows "Above sensitivity? NO" and "Shannon limit? NO" with 3.3 bps, while the text claims a 15 dB margin at 10 kbps: *confirmed and explained*. A bandwidth of B= 0.125 Hz was used for 0.125 MHz, which flips C/N by 60 dB, and the 15 dB claim reproduces only with the Rx gain corrected to 24.2 dBi and the wrong EIRP kept (B-02, B-05).
5. The thermal case labels: the 1-node "best case" is case 3 (gold polyimide, $e^* = 0.005$, 7.9 W), not case 5, and it refers to a 1.6×1.06×0.96 m box, not to the penetrator (A-02).

New critical/high findings. The audit also produced new findings. Table 34's heater powers are reproduced by a one-node battery-only model (Section 5.3). The report's flare would raise the peak deceleration 4–10× (Section 5.2). Finally, a Titania target is reachable, which contradicts the "820 m/s" row, because the report used the 2045 fast flyby rather than the 0.47 km/s flyby shown in its own Fig. 14 (A-13).

4 Requirements and Concept of Operations (v2)

Inherited, unchanged: the requirements carried over from the 2025 study are COTS/TRL 6+ components, the SIMPLEx guidelines, ≤ 2 W drawn from the orbiter in cruise, do no harm to the orbiter, UHF communication to the flagship, and a mass of ≤ 180 kg. **Revised (v2):** the requirements changed for v2 are

- SYS_SD_3 (ΔV capability): ≥ 330 m/s is required at MEV mass (it was 450 m/s); the capability is 400 m/s with four STAR 5D motors at 149 kg (Section 5.1).
- New: P(acceptable impact) ≥ 0.9 for every plausible flyby orientation. An impact is *acceptable* when it is a hit, the incidence is ≤ 10°, the impact speed is ≤ 0.95 km/s, the flagship is ≥ 5° above the local horizon at impact+30 min, and the slant range is ≤ 7,000 km. These criteria are ASSUMED and traceable to the structural model.
- SYS_SD_1 (survive 30 min): this is now demonstrated with a 0 W heater. A 2 h hold at ≤ 20 W is met for ∼ 88% of the sampled thermal parameter space, and 6 h is kept as the stretch objective.
- Link: the rate is capped at 6 kbps (the UOP probe relay [1]) unless the UOP UHF G/T is confirmed to be higher. The memory is sized to the link (1 GB class), not to 50 GB.
- Decadal traceability: Q5.1c (deceleration/strength profile), Q8.2f (bulk thermal conductivity) and Q12.8a (partial: descent imagery) are kept, while Q5.1a (interior structure) and Q4.4b (plume composition) are descoped (Section 5.6).

CONOPS (v2). The v2 concept of operations runs as follows. The penetrator is released at T−4.4 h and spun up, and it burns along the ΔV vector with the SRB nozzles aft and the nose forward. It is then despun and performs a slow (≤ 0.15°/s) slew of ≈67° to the velocity-tracking attitude, which is completed in ≈450 s on 0.06 N m s wheels. A coast of 4.6 h follows, with UHF telemetry at the rate the link supports, and the core is preheated during the last hour. Impact occurs ≈12 min after the flagship's closest approach. The accelerometer records the impact pulse, the RTD decay and the seismometer are recorded for ≥30 min, and the split aftbody carries the antenna.

5 Results

This section applies the models described in Section 2 to the delivery, impact, thermal, power, link and science problems in turn, and then tests the novel design ideas against the same baseline.

5.1 Trajectory, Release and Delivery

5.1.1 Model

The delivery is modelled with planar circular-restricted three-body dynamics (a Uranus point mass and an Umbriel point mass on a circular orbit, with the indirect term included), integrated with an 8th-order Dormand–Prince scheme (rtol 10^{-11}). The flagship state is fixed by the UOP flyby table (altitude and relative speed at closest approach [1]). The table does not give the orientation of the flyby, so all four combinations of (flagship outside/inside the moon's orbit radius) $\times$ (prograde/retrograde relative velocity) are run and reported as a spread. We treat the burn as instantaneous, since a STAR 5D burns for 3.03 s [23]; the finite-burn, thrust-offset and spin effects are carried in the dispersion model. Central hits (zero impact parameter, incidence $< 2°$) are found by shooting on the signed aim parameter $g = h/(R\,v)$, and minimum-ΔV solutions by projected Gauss–Newton from several deterministic starts. Surface co-rotation (10 m/s) is included. The Uranus tide also matters: at 8,000 km it displaces the trajectory by $\sim$10%.

Validation against the report. For the Umbriel row of Table 2 (release at 4.4 h, $\Delta V = 415$ m/s), the independent central-hit solution gives 401 m/s $\rightarrow$ 0.86 km/s and a flight time of 3.76 h, against the report's 901 m/s, 4.2 h and 86°. This is within 4% in speed and 10% in time. The Ariel, Oberon and Titania rows are not reproduced, because the objective used to generate them is not stated (A-12).

5.1.2 Minimum ΔV versus Release Time

Figure 2 shows the minimum retarget ΔV and the matching impact speed as functions of the release time.

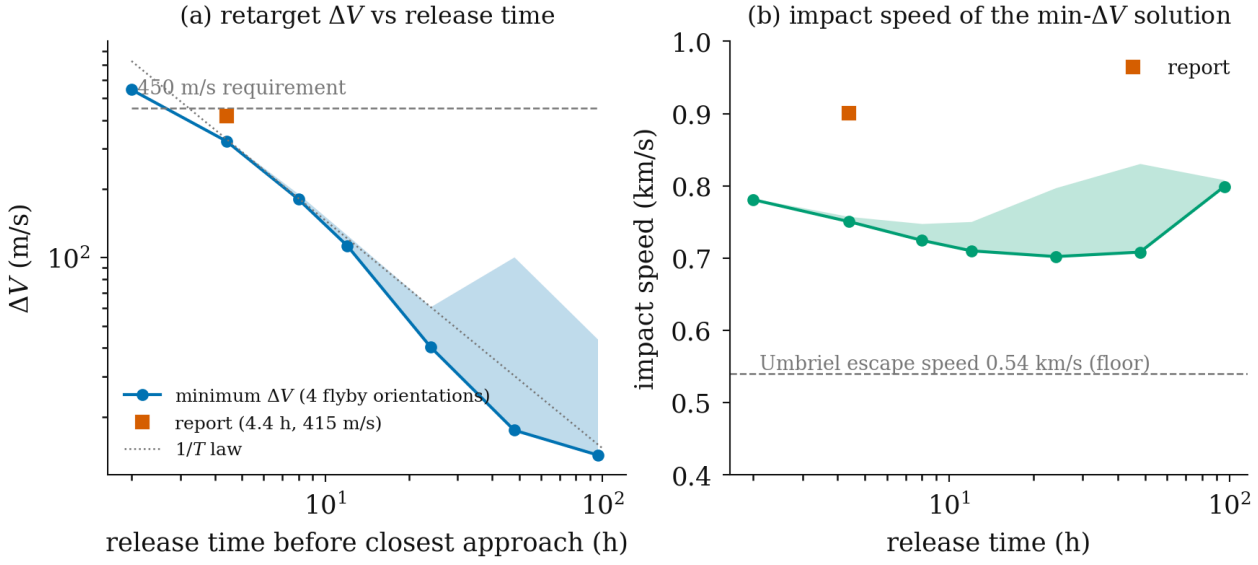


Figure 2: (a) Minimum retarget ΔV for a central hit on Umbriel versus release time; the band covers the four unknown flyby orientations. Below $\sim$12 h the law is $\propto 1/T$ (a straight-line retarget of a $\sim$5,200 km miss), while beyond $\sim 1/n = 16$ h orbital dynamics flatten it and the multi-start solver becomes unreliable, so the 48–96 h values should be treated as order-of-magnitude. (b) Impact speed of the minimum-ΔV solution, which is 0.70–0.83 km/s; the floor is Umbriel's escape speed of 0.54 km/s.

At the report's release time the minimum is 323–328 m/s (not 415, and not 450), with $v_{\mathrm{imp}} = 0.75$–0.76 km/s. Releasing 12 h earlier cuts the requirement to 121–125 m/s, and releasing 24 h earlier cuts it to 40–61 m/s (57–73 m/s for line-of-sight-feasible solutions). A burn at closest approach is the *worst* choice (855 m/s at 6 min before closest approach). The reason is that the Oberth factor available at this distance is only 1.05, because Umbriel's escape speed at 5,939 km (0.17 km/s) is far below the 0.52 km/s flyby speed.

5.1.3 Impact Speed, Line of Sight and the Pareto Front

Figure 3 shows all central-hit solutions in the impact-speed versus ΔV plane, together with the encounter geometry of two designs. With the line-of-sight constraint, the lowest impact speed reachable with $\Delta V \leq$

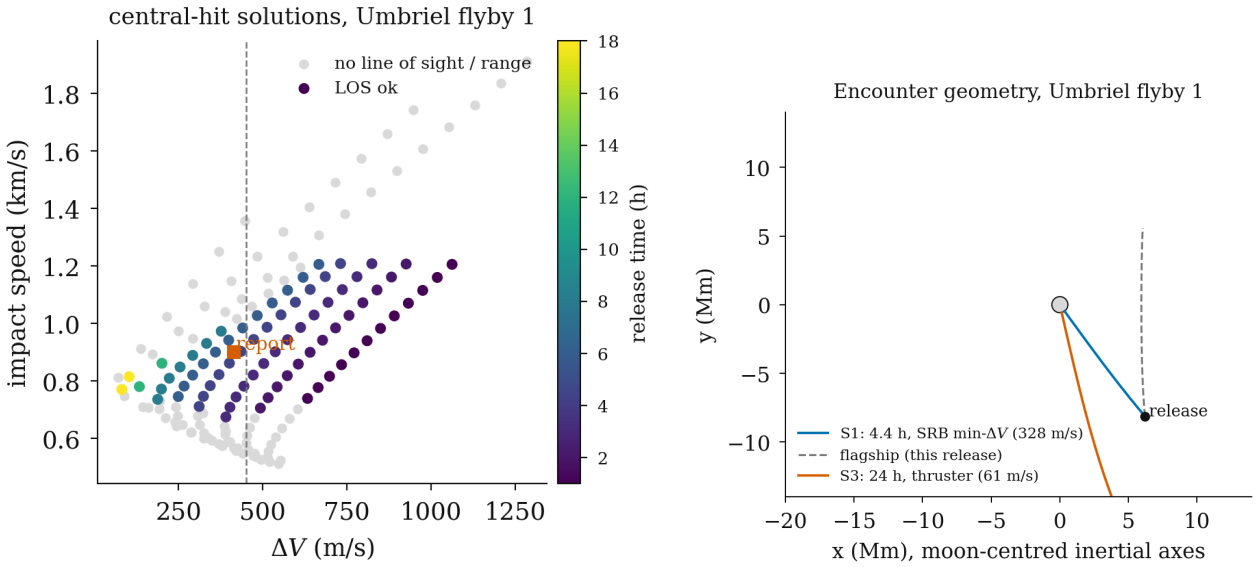


Figure 3: Left: all central-hit solutions (release 1–24 h) for Umbriel flyby 1, impact speed versus ΔV; the dark dots keep the flagship $\geq 5°$ above the horizon and within 7,000 km at impact+30 min. Right: encounter geometry for the 4.4 h SRB and 24 h thruster designs.

300/350/400 m/s is 0.74/0.71/0.67 km/s (impact 0.15–1.5 h after closest approach); without it, the values are 0.62/0.59/0.57. The constraint therefore costs 0.1 km/s, because the flagship is leaving and the link range grows.

5.1.4 Dispersion and Probability of Acceptable Impact

Figure 4 shows how the probability of an acceptable impact falls with each error source. The error budget

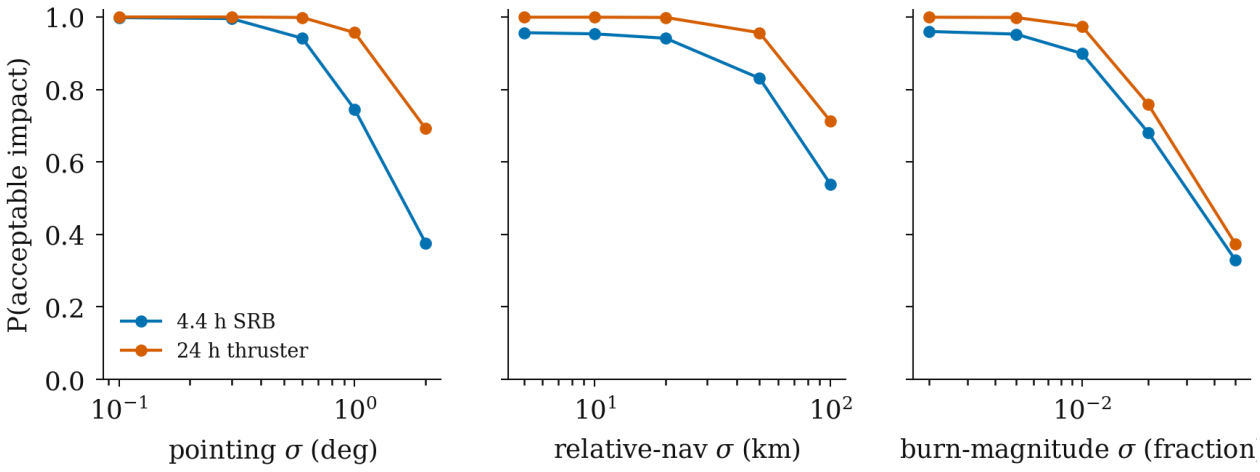


Figure 4: P(acceptable impact) versus pointing, relative-navigation and burn-magnitude error for the 4.4 h SRB design (S1) and a 24 h thruster design (S3), flyby orientation (+,+).

(1σ, ASSUMED) is: burn magnitude 0.7% (SRB) or 0.5% (thruster); pointing 0.6° per axis (the thrust offset after spin stabilisation, B-30); release timing 5 s; relative-navigation position 20 km (Uranian-moon ephemeris plus flagship OD); and separation velocity 1 cm/s. The nonlinear encounter is linearised by finite differences and checked against direct integration, and the Monte Carlo uses $N = 40{,}000$ seeded samples. For the re-simulated report concept P(acceptable) is 0.89, and for the minimum-ΔV SRB design it is 0.94, in all four flyby orientations. The dominant terms are the burn magnitude (for which ∂g is largest) and the

pointing. With 1° pointing P falls to 0.75, and with 2° to 0.38; with 50/100 km navigation error it falls to 0.83/0.54, and with 2% magnitude error to 0.68. These results define requirements that the report lacks: pointing $\leq 0.6°$ (1σ) including the thrust-vector offset, navigation ≤ 20 km, and impulse repeatability $\leq 1\%$.

Line of sight. In 4 of 4 orientations the 4.4 h design keeps the flagship $\geq 5°$ above the horizon with probability ≥ 0.99. The earlier-release designs do not: P(line of sight) is 0.75–0.87 at 12 h, 0.58–1.0 at 24 h and 0.0–1.0 at 48 h. The report's claim of line of sight "because flight time < time before closest approach" is not a valid argument (B-11).

5.2 Impact, Penetration and Structure

5.2.1 Model

The penetration is modelled with dynamic spherical cavity expansion for an ogive nose of caliber-radius-head ψ and radius a:

$$F(z, v) = f_{\text{nose}}(z)\,\pi a^2\left[R_t(z) + N\rho_t(z)\,v^2\right] + F_{\text{friction}}, \qquad N = \frac{8\psi - 1}{24\psi^2}, \qquad R_t = \tfrac{2}{3}Y\left[1 + \ln\frac{E}{3Y}\right], \quad (1)$$

where f_{nose} is the nose factor, R_t the target resistance, N the nose shape factor, ρ_t the target density, Y the target yield strength and E its Young's modulus. The force is integrated as $m\,dv/dt = -F$. For the report's CRH-5 nose $N = 0.065$, whereas the report's Tate–Alekseevski formula uses a blunt face ($N = 0.5$) and so over-predicts the inertial term 7.7×. We reproduce the report's 10.4 MN only with $R_t = 0.6$ GPa (the 0.454 GPa from its own eq. B.5.3.4 gives 8.8 MN), and we reproduce its second model's 5.3 MN. Three target classes are used, with ASSUMED prior weights and results shown per class: *hard ice* (Y = 20–150 MPa log-uniform, rate enhancement 1–2.5, E = 9–12 GPa), *regolith over ice* (0.5–15 m of compacted regolith, R = 1.5–24 MPa, over hard ice), and *porous regolith* (R = 0.2–8 MPa, ρ = 400–900). The penetrator mass is 110–150 kg (CBE to MEV), $a = 0.058$ m and the friction is 0.02–0.2; the inputs are Sobol-sampled with $N = 2048$.

5.2.2 Results

Figure 5 shows the deceleration pulses and the depth distributions for the three target classes. At 740 m/s

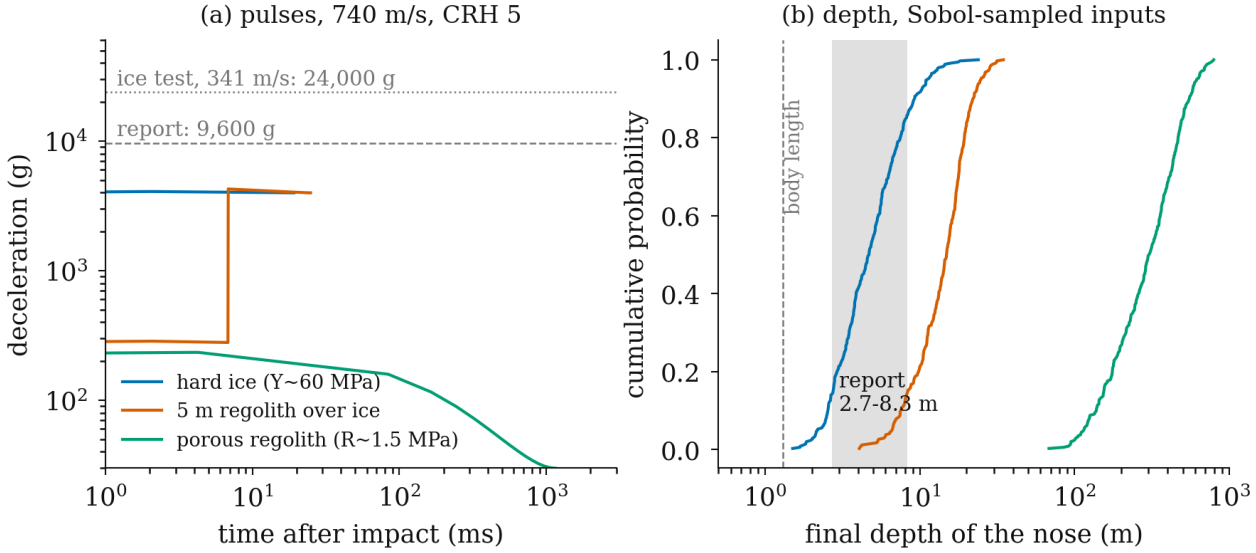


Figure 5: Left: deceleration pulses at 740 m/s for the median members of the three target classes. Right: depth distributions; the shaded band is the report's 2.7–8.3 m.

the *hard-ice* class gives a depth of 2.2–11.7 m (5–95%) and a peak deceleration of 3–18 kg. The *regolith over ice* class gives 6–24 m, and the *porous regolith* class 121–596 m at 143–357 g. Averaged over the three classes, 11% of cases exceed 10,000 g. The report's 5.4–10.4 MN and 3.4–8.3 m are therefore the stiff-ice corner of these distributions. For comparison, the empirical Sandia equation (warm ice, $S = 4.5$) predicts 39 m at 900 m/s for 119 kg with a CRH 5 nose [14].

Nose shape. In hard ice the peak deceleration is strength-dominated and nearly independent of ψ (4.4–4.1 kg from $\psi = 3$ to 12). In porous regolith it falls from 1,600 g (hemisphere) to 120 g ($\psi = 12$), while the depth rises from 70 to 435 m. A sharper nose is therefore not a free improvement, since it trades shock for burial.

Flare. A drag flare of radius 0.12–0.2 m raises the hard-ice peak deceleration from 4.2 kg to 16–42 kg and still leaves the nose 2–3 m deep, which is more than the 1.3 m body, so the antenna is buried anyway. The heritage solution is a split forebody/aftbody joined by a cable, with the antenna on the aftbody at the surface (DS-2, Mars 96).

Structure and instruments. The required ring wall (Ti-6Al-4V, 0.88 GPa yield, factor of safety 1.25, 100 kg behind the section) is 1.9/3.8/7.6/19/38 mm at 500/1,000/2,000/5,000/10,000 g. The allowable incidence for an 8 mm wall is ∼11° at 1,000 g but only 3.6° at 3,000 g (lateral load $F \sin\alpha$, $\kappa = 1$, ASSUMED). The 10° acceptance criterion of Section 5.1 is therefore generous for the hard-ice case, and it demands the 0.6° pointing given above. For instrument shock, an AC-coupled accelerometer chain with a 4 Hz corner under-reads the velocity from a 10 m pulse by 27% (5–9% after deconvolution with a ±25% corner error). The Artius broadband seismometer has no shock rating in the report, and the only demonstrated seismometer survival is 8,000–10,000 g (Lunar-A) [20].

Oblique impact or tumbling to cut peak g fails, because the normal-velocity gain is small compared with the lateral bending load at the allowable incidence (Section 5.7).

5.3 Thermal

5.3.1 What Table 34 Actually Says

The report's 70-node model is a cuboid. Its ice cases hold the battery at a constant heat load and ask when it crosses −55°C. We find that a one-node model, $C\,dT/dt = P - G(T - T_{\mathrm{ice}})$, where C is the heat capacity, P the heater power and G the conductance to the ice, reproduces Table 34 with 3.4 W rms for $G = 0.92$ W/K and $C = 6.2$ kJ/K (battery only). The steady states (148 W at 70 K and 185 W at 30 K) give $G = 0.917$ and 0.918 W/K, which is the 1 W/K mount assumption (10 W/m^2/K × 0.1 m^2). Two assumptions therefore set the 140 W: the structure sits at ice temperature instantly, and only 2.6–6 kJ/K of thermal mass is inside the insulation.

5.3.2 Improved Model

In the improved model the core (batteries, electronics and instruments; $C = 6$–28 kJ/K) is connected through a conductance G_{ins} to a shell ($C_s = 45$ kJ/K), which loses heat by radial finite-volume conduction into crystalline ice ($k = 567/T$, $c = 7.49T + 90$ [24]) or porous regolith ($k = 0.01$–0.5 W/m/K). The conduction is radial 1-D with a geometric end-factor of 1.3 (ASSUMED). A full 2-D axisymmetric (r,z) model, the impact-heated sheath and shock-surviving insulation are not modelled (Section 7.4). Figure 6 compares the one-node fit with Table 34 and shows the heater power and the time history from the improved model.

5.3.3 Results

Even at the report's $G = 0.92$ W/K, a core with 11 kJ/K starting at −20°C holds −40°C for 30 min with *no* heater (0.67 h unheated at 70 K). Starting from the free-flyer equilibrium of −39°C with a −50°C limit, it needs 12.6 W. At $G = 0.25$ W/K a 2 h hold needs 11.5 W (cold start), and the unheated core survives 2.0 h in 70 K ice and 7.1 h in porous regolith ($k = 0.05$). A Sobol Monte Carlo of 512 samples (over $G = 0.08$–0.6 W/K, C, ice temperature 50–90 K, ice versus porous target, start temperature −39 to −10°C, and dissipation 3–6 W) shows that a 2 h hold needs no heater at all in 71% of cases, ≤24 W in 90%, and ≤39 W in 95%. Figure 7 shows the heater power against hold duration and the Monte Carlo distribution.

Answer to the 30-minute question. A 30-minute survival is achievable on 8 LSH20-class cells (about 1 kg) rather than 4 kg of commercial cells (Section 5.4). The *trade between heater power and duration* is shown in Figure 7: a hold of 6 h needs 0–26 W at $G \le 0.25$ and rises to 131 W at the report's $G = 0.92$. Achieving $G_{\mathrm{ins}} \le 0.25$ W/K requires about 10 mm of aerogel-class insulation around a 0.24 m^2 core, plus Ti standoffs. The vacuum conductivity of aerogel (0.004–0.009 W/m/K) is UNVERIFIED here, and its shock survival is a TRL risk.

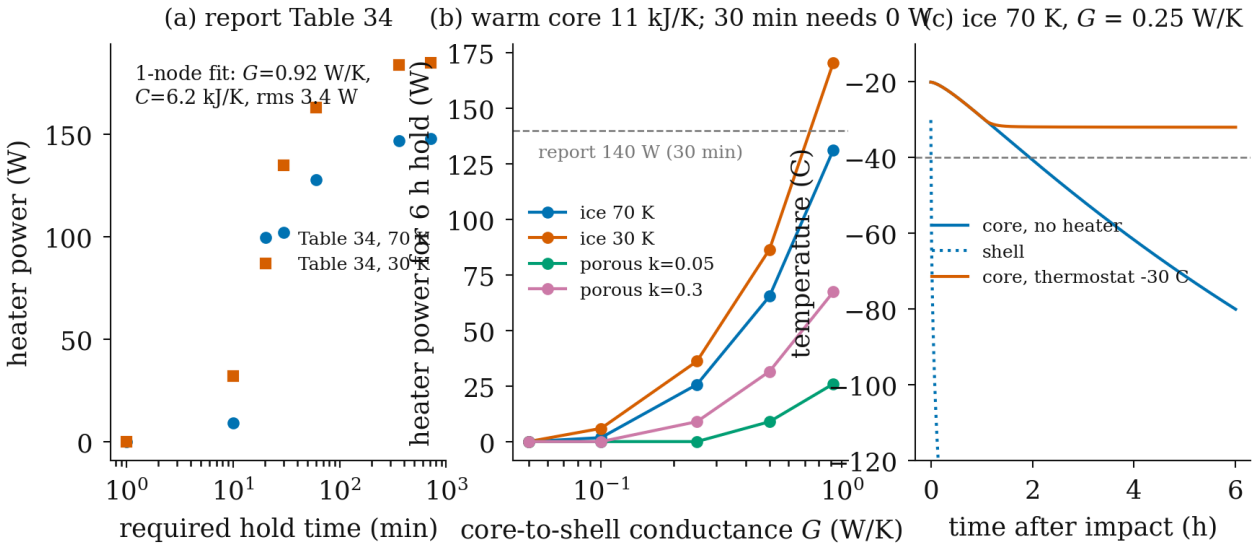


Figure 6: (a) Report Table 34 and the one-node fit. (b) Heater power for a 6 h hold versus core-to-shell conductance; 30 min needs 0 W everywhere on this plot. (c) Time history for $G = 0.25$ W/K in ice at 70 K.

5.4 Power, Mass and Volume

Mass. Itemising the percentage split gives a wet CBE of 160.2 kg and a wet MEV (AIAA S-120 contingency by maturity [25]) of 191 kg, or 163 kg dry. The 180 kg limit is a dry limit (A.3), so the system fits only if the dry interpretation is confirmed, and even then only without system-level margin (the dry MEV plus 15% is 187 kg). The v2 design saves the 2–3 kg of battery and the reaction-wheel/spring flip hardware, but it stays above 180 kg wet. This is an open requirement-interpretation issue. Figure 8 shows the itemised mass and the survival time of the report's cell stack.

Li-SOCl$_2$ cells at cold, high current. The cell models (capacity versus temperature and current, internal resistance rising at low temperature, and voltage delay after 13.4 years) are fitted to the manufacturer plots, with an RMS error of 0.6–0.7 Ah that lies inside the ±1 Ah digitisation band. A 4 kg stack of bobbin or spiral commercial cells cannot deliver the report's 140 W at ≤ 253 K (it browns out). Only a DS-2-class cell scaled with a 5× or 20× bracket survives (150–243 min), and that result rests on a single datum [22]. The report's 4s48p topology passes 30 min at 233 K under the DS-2-class assumption. The report's total energy (195.9 Wh, against ≈1.3 kWh available in 4 kg) is not the constraint; the *pulse current at low temperature* is. With the v2 thermal design the load is ≤20 W in a warm core, so 8 LSH20 cells (1.0 kg) suffice for 30 min at 20 W and 253 K; v2 allocates 2 kg.

v2 energy budget (ASSUMED loads). The cruise load is 0.64 W. The 4.6 h descent draws CDH 2 W, ADCS 2.1 W and comms 1 W (20% duty), for 23.5 Wh. The imagers add 10 W for the last 30 min (5 Wh), and the core preheat 10 W for the last hour (10 Wh). Post-impact, 2 h at 13.8 W (CDH, seismometer, comms and a 10 W average heater) uses 27.6 Wh. The total is ≈66 Wh, against the report's 195.9 Wh.

5.5 Communications and Data

The report's link is for a 2 GHz patch to a 1 m orbiter antenna, whereas UOP carries X-band, Ka-band and a UHF low-gain antenna [1]. We therefore rebuilt the budget at 401.585625 MHz (the Mars Proximity-1 return band [28]) with rate-1/2 convolutional coding (required E_b/N_0 of 4.4 dB, ASSUMED), a transmit gain of −3 to +3 dBi (a 0.15 m patch is only 0.2 λ at 401 MHz), an orbiter gain of 0–6 dBi, a system noise temperature of 300–800 K, and burial of 3–30 m with $\tan\delta = 1.6 \times 10^{-4}$–$1.6 \times 10^{-3}$ (plus a 5×10^{-3} tail, since no measured value at 100 K was found). Burial loss in dB per metre scales with frequency, so UHF loses five times fewer dB per metre than 2 GHz. The report's argument for 2 GHz is therefore backwards. Figure 9 shows the probability of closing the link against range and the data volume returned.

Results (Monte Carlo, N = 20,000, fixed seed). At 5.4–6.5 Mm the median 10 kbps margin is -7.5 dB (5–95%: −13.7 to −1.9 dB). P(margin ≥ 3 dB) is 0.4% at 6 kbps and 0% at 10 and 20.1 kbps. At 20.1 kbps the link loses its 3 dB margin at a median range of 1.7 Mm (0.9–3.0 Mm). The total data returned over the descent and 30 min is ≈16 MB (9–25 MB), of which only 0.15 MB (0.04–0.55 MB) is returned after impact. The report's 20 GB (and 50 GB memory) is 10^3–$10^5\times$ what can be returned. Reproducing "15 dB at

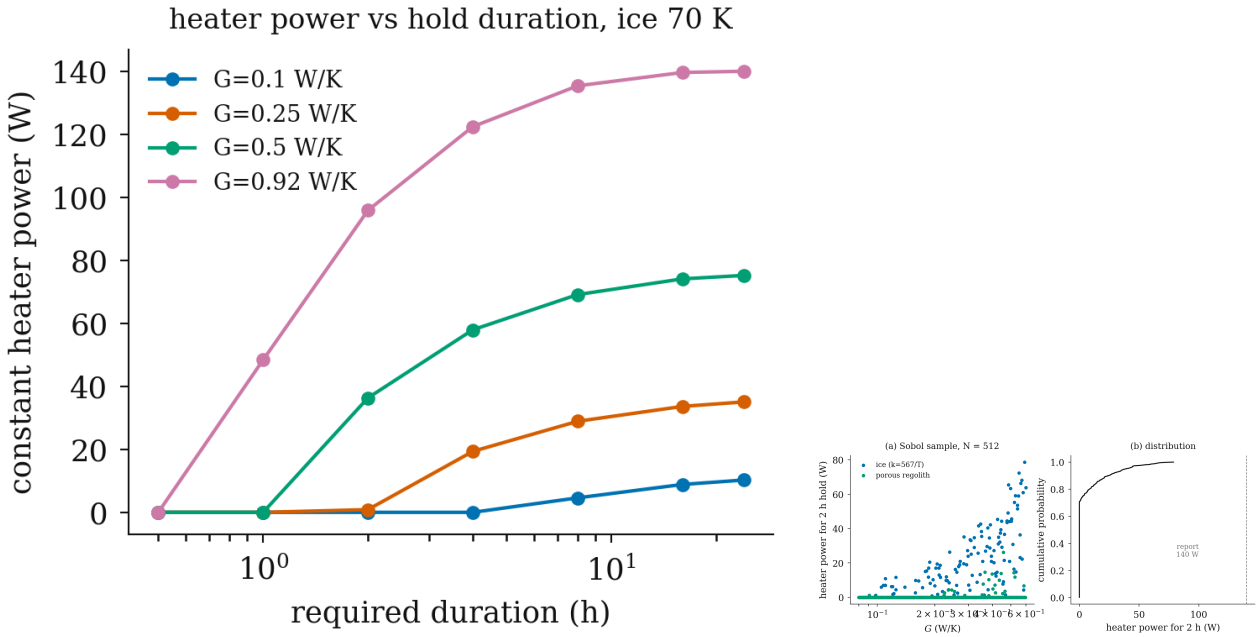


Figure 7: Left: constant heater power needed to hold the core above −40°C versus duration, for four insulation levels (ice at 70 K). Right: Monte Carlo distribution for a 2 h hold.

10 kbps at 6.5 Mm" would need an orbiter dish of 1.4 m at 2 GHz or 4.4 m at UHF (free space). *These results are dominated by the assumed antenna gains and noise temperatures. They identify the UOP UHF G/T as a required input, not a conclusion about the final system.* Choosing the second Umbriel flyby (3,653 km) shortens the range by about 1.5 Mm (+2.6 dB), at the price of a 0.89 km/s impact speed.

Store-and-forward. The data are sent in priority order (volumes in MB): accelerometer window 0.02, RTD 0.043, housekeeping 0.045, seismometer 0.65, and thumbnails 0.1 each. Only the first three fit the post-impact volume at the median.

5.6 Science Return

Nine traceability rows were built against the Decadal questions Q12.8a, Q4.4b, Q5.1c, Q5.1a and Q8.2f, each with a measurement model and an achievable uncertainty (`analysis/out/science_traceability.json`). Table 1 summarises them, and Figure 10 shows the seismic signal-to-noise ratio and the thermal-conductivity uncertainty behind the verdicts.

Table 1: Science Traceability Summary

Question	Measurement	Limit	Verdict
Q5.1c	flow stress / resistance $R(z)$	accelerometer + depth: 9% (median) to 27% (p95) on R, 13% on Y excluding model error	achievable
Q8.2f	bulk thermal conductivity	heated-cylinder transient, Cernox ±16 mK: σ_k/k 7–23% at 30 min	achievable
Q12.8a	surface imagery	only the last ∼300 s (<70 m GSD) cannot be sent live; best live image GSD ∼110 m (median)	partial
Q5.1c	porosity, coda attenuation	single station, near-surface only	partial
Q4.4b	plume/ejecta composition	plume observability from 5.4–6.5 Mm; spectrometer on UOP UNVERIFIED	not achievable
Q5.1a	interior structure, Crary waves, natural seismicity	single 30-min station; direct impact-wave range 17/46/316 km (5/50/95%); $P(\geq 1$ natural event in 30 min)= 6%	not achievable

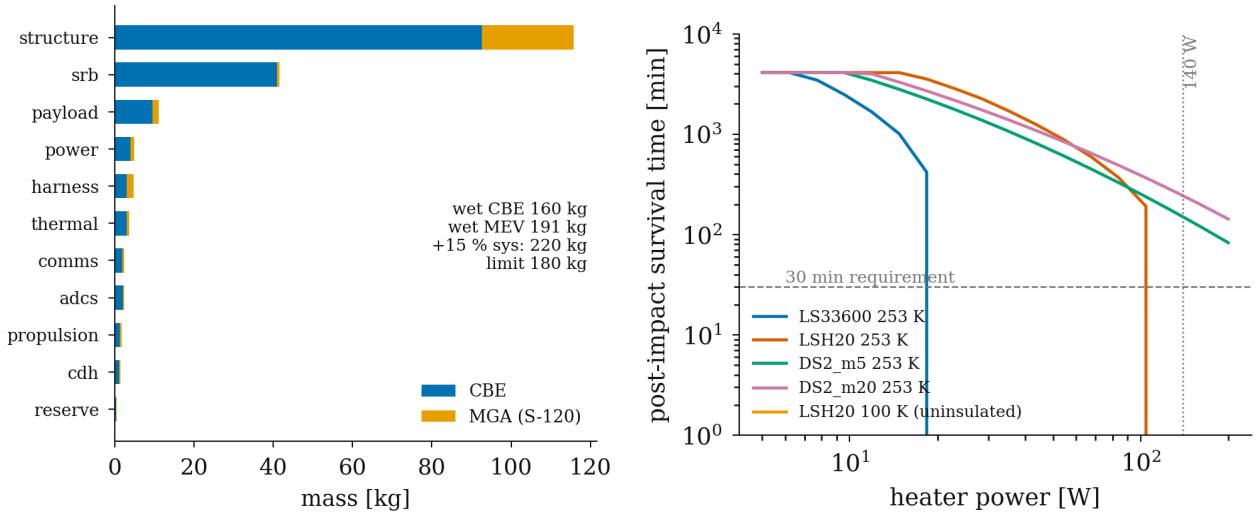


Figure 8: Left: itemised mass with contingency by maturity. Right: survival time versus heater power for a 4 kg cell stack, with cell models fitted to Saft LS33600 and LSH20 data [26, 27].

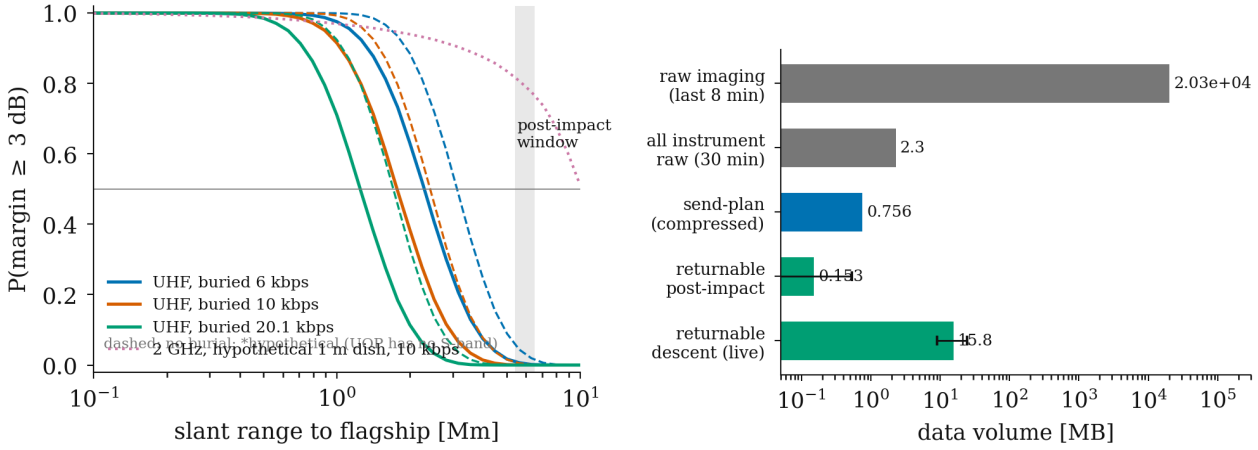


Figure 9: Left: P(margin $\geq$ 3 dB) versus range for the UHF link. Right: data volume returned versus data volume produced.

5.7 Novel-Methods Workstream

Each idea is simulated against the re-simulated report baseline (S0: 4.4 h, 401 m/s, 0.86 km/s, P(acceptable) 0.89) with the same machinery. The objectives are stated rather than hidden: minimise the impact speed v_{imp}; minimise ΔV and the propulsion mass it implies; maximise P(acceptable impact); and maximise science per kg. A strict literature check was also made. A flyby-released probe retargeted by its own burn is already in the literature (Wurz et al. 2017, with release 27 h before closest approach at Europa and ΔV of 0.5–0.9 km/s [29]; Sun et al. 2020 for Enceladus [30]; Davis 1976 for the error analysis of orbiter-launched penetrators [31]), as is the hypervelocity penetrator concept (Winglee et al., Luna-Glob, $>$ 1.5 km/s [32]). We found no Uranus-moon penetrator study and no study specific to the UOP tour.

Figure 11 compares the release strategies, Table 2 lists the candidate designs, Table 3 gives the full scoreboard, and Table 4 the alternative target moons.

What failed, and why. Burning at periapsis fails because there is no Oberth benefit at this distance. Decelerating before impact fails because, in stiff targets, the peak deceleration is set by target strength rather than speed. Oblique impact fails on bending, and a network of seismic penetrators does not fit within the mass cap. **What the numbers do not support claiming.** The numbers do not support claiming that a lower impact speed lowers shock (it does not in hard ice), that early release is robust (it is not, across flyby orientations), or that the Titania gravity-assist bound is a design.

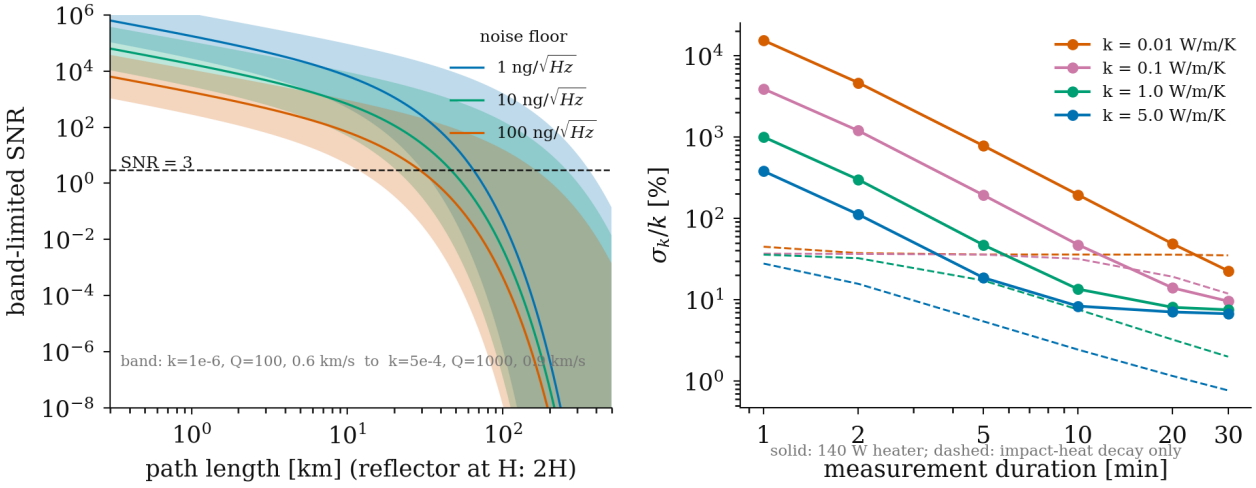


Figure 10: Left: seismic SNR of the penetrator's own impact versus distance for source efficiencies of 10^{-6}–5×10^{-4}. Right: thermal-conductivity uncertainty from the heated-body transient.

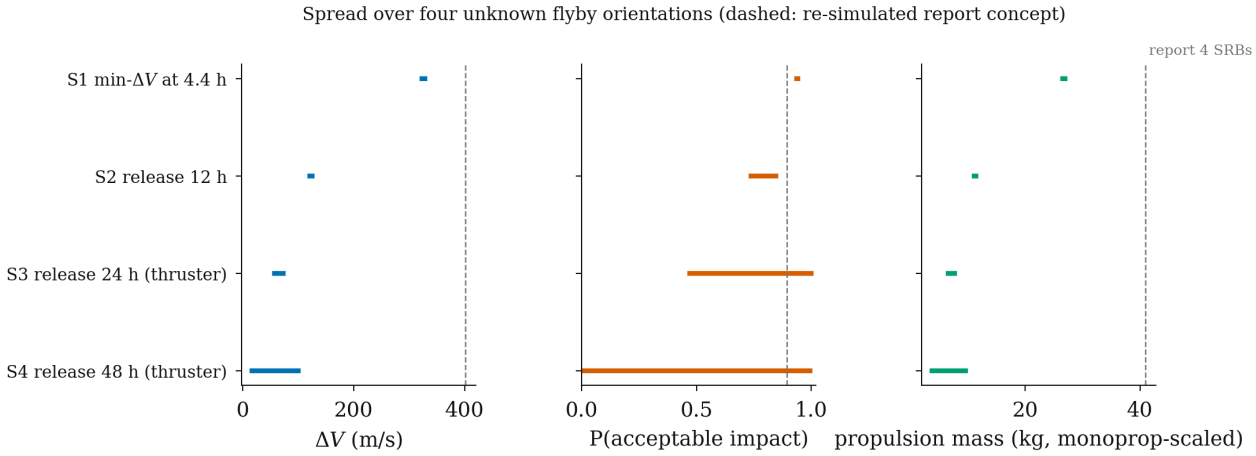


Figure 11: Spread of ΔV, P(acceptable impact) and propulsion mass over the four unknown flyby orientations for the candidate release strategies (dashed: re-simulated report concept; right-hand dashed: the report's 40.9 kg SRB stack).

6 Mission-Level Analysis and Comparison with Practice

The models of Section 5 answer whether the penetrator can be delivered, survive and talk. A mission design also needs the elements that SMAD lists for any concept of operations [3, pp. 15, 30–31]: a timeline from launch to the end of the mission, a communications and data architecture, a reliability argument, a cost class, a verification flow and, for a body that may hold water, planetary protection. This section adds these elements and compares each with what flown missions actually did. The numbers are produced by `analysis/perci/v3_mission.py`; external values carry their sources and are tagged as in Section 2.

6.1 Comparison with Flown and Studied Penetrators

We first place PERCI among the penetrators that have been built, tested or studied. Table 5 lists them, and Figure 12 shows their mass and impact speed together with impactors and landers.

Every survivable penetrator in the record weighs 3.6–45 kg and hits at 80–341 m/s. PERCI v2 hits at 750 m/s with 119–149 kg, which is 2.2 times the speed of the ESA ice sled test (20 kg, 341 m/s, 24,000 g [13]) and 29 times its kinetic energy (33.5 MJ against 1.2 MJ). Moreover, the flight record of the class is short. The two DS-2 probes were lost with no probable cause identified [33], the Mars 96 penetrators were lost with their launcher [15], Lunar-A was cancelled in 2007 after its penetrator development overran [36], and MoonLITE stopped after Phase A [38]. No penetrator has yet returned data from another body. The designs that were carried furthest share three features that PERCI lacks: a mass of about 13 kg with about 2 kg of science [38, 40], an orbiter that revisits the site every few weeks for relay [35, 38], and an impact load

Table 2: Candidate Designs (Central Hits, Umbriel Flyby 1). Propulsion mass is SRB-scaled (S0–S2) or hydrazine-monoprop-scaled (S3–S4, Isp 220 s ASSUMED, tankage 25%, 2.5 kg hardware). The ΔV, impact speed and P ranges are over the four unknown flyby orientations.

Design	ΔV (m/s)	v_{imp} (km/s)	P(acceptable)	prop. mass (kg)	orient. tested
S0 report-like (4.4 h, 415 m/s)	401–401	0.86–0.86	0.89–0.89	33–33	1
S1 min-dV at 4.4 h (SRB)	323–328	0.75–0.76	0.94–0.94	26–26	4
S2 release 12 h (small SRB)	121–125	0.73–0.75	0.74–0.85	9–9	4
S3 release 24 h (thruster)	57–73	0.71–0.80	0.47–1.00	7–8	4
S4 release 48 h (thruster)	17–100	0.68–0.81	0.00–0.99	4–10	4

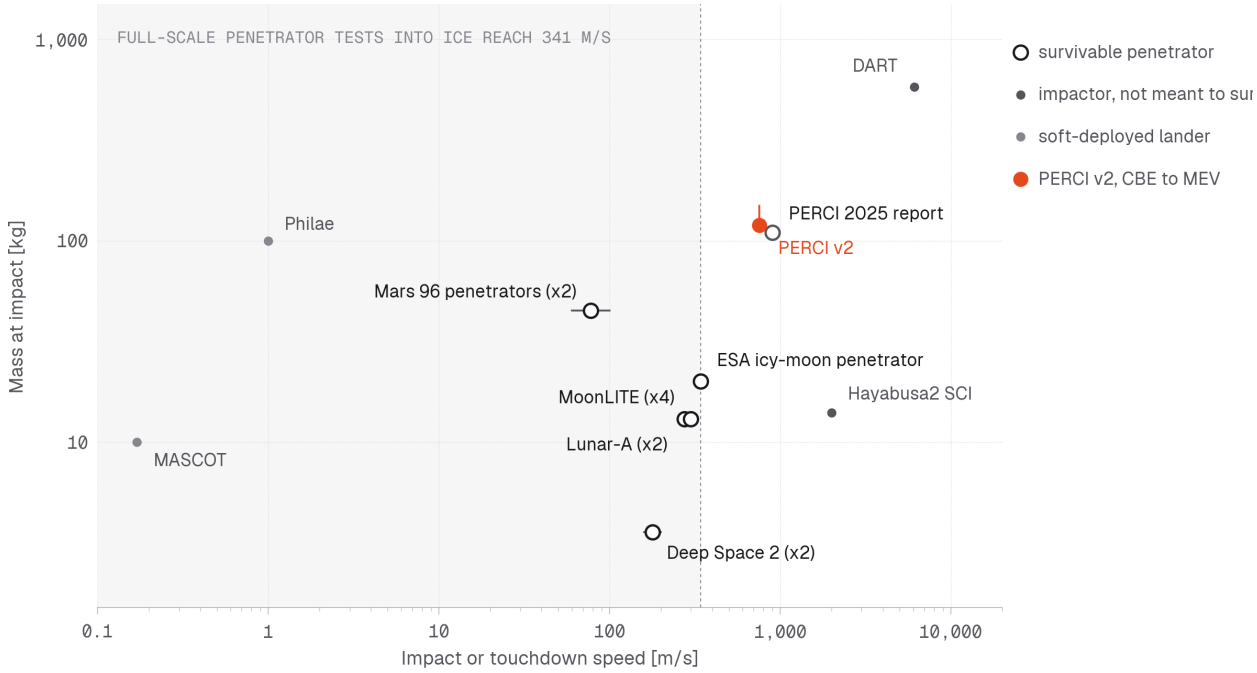


Figure 12: Mass at impact versus impact speed for flown, tested and studied penetrators, impactors and landers. PERCI v2 is 2.2 times faster than the fastest full-scale penetrator test into ice and 2.6 times heavier than any survivable penetrator built. Sources in Table 5; DART and SCI from [41, 42], landers from [43, 44].

that was demonstrated in a full-scale test [13, 45]. Section 6.8 tests whether a 13 kg-class body would also work at Umbriel.

6.2 Deployment from a Flagship

To connect the release of Section 4 to practice, we compare it with how secondary payloads have been released from carriers (Table 6).

Four lessons apply directly. First, the flagship should give the spin. Huygens was released by a Spin/Eject Device that left the probe at 0.35 m/s and 7.5 rpm, and about 30 kg of support equipment stayed on the orbiter [46, 47]. PERCI instead carries its own spin-up motors and yo-yo despin (1.75 kg MEV). A flagship-side spin table would move that mass to UOP and remove two pyrotechnic events from the penetrator. Second, the penetrator must be testable after integration. The DS-2 board found that the probes could not be functionally tested after aeroshell integration or before separation, and called this inappropriate [33, Sec. 3.3.2]; Philae was released although its cold-gas thruster had failed its pre-separation check [51]. PERCI therefore needs an umbilical checkout and a written go/no-go rule for the T−30 h timer load. Third, the release command sequence is itself a single point: MINERVA was released at about 200 m while the spacecraft was climbing, not at the planned 70 m, and escaped [54]. Fourth, thermal state at release matters. The Artemis I CubeSats needed a 1 rpm thermal roll of the upper stage, or their batteries would have frozen, and only 5 of 10 achieved partial or full success [55].

Figure 13 shows what these releases mean for PERCI. With a 1 km stand-off before the SRBs fire (an



Figure 13: Distance from the flagship at SRB ignition versus delay for three flown spring separations (a), and P(acceptable impact) of the S1 design versus separation-velocity error (b). A Huygens-type release needs 48 min for a 1 km stand-off, and P(acceptable) stays at 0.94 for any error up to 15 cm/s. The 1 km keep-out is ASSUMED; separation speeds from Table 6.

ASSUMED value; the report gives none), a 0.35 m/s release must happen 48 min before the burn, a 0.19 m/s release 89 min before and a 1.1 m/s release 15 min before. The separation therefore moves to about T−5.2 h. Philae's delivery was reconstructed at 0.174 m/s against a 0.187 m/s design [49, 50], a 7% knowledge error. Applying 7% to each spring and rerunning the dispersion Monte Carlo of Section 5.1 leaves P(acceptable) at 0.94 for all cases up to 15 cm/s, so the spring is not an aim driver; pointing and burn repeatability remain the drivers. The flagship constraints are different. UOP must not fire thrusters during the SRB burn, it must hold its UHF antenna on PERCI from closest approach plus 12 min to plus 42 min, and that window falls inside its own closest-approach science pass at Umbriel. This conflict has to be negotiated with the UOP science team, and it is the main operational cost that PERCI imposes on the flagship.

6.3 Mission Operations Timeline

Figure 14 gives the timeline from launch to relay, as SMAD asks for in the concept of operations [3, pp. 30–31] and the mission operations plan [3, pp. 588–589].

The UOP baseline launches on 9 June 2031, inserts into Uranus orbit on 5 December 2044 after a 13.5-year cruise, and flies past Umbriel on 22 June 2047 [1]. The penetrator is therefore stored for 16.0 years between launch and use, not 13.4: the 2025 report counted only the cruise. During that time it needs periodic checkouts through the umbilical (we ASSUME one per year and one after orbit insertion), and it draws its cruise power from the flagship. The 2026 update of the UOP study states that a 2031–2032 launch is precluded by current funding, moves launch readiness to 2035–2040 and drops to two next-generation RTGs with 350 W at the end of the mission [4]. The 2047 dates above are therefore placeholders, but the storage time does not shrink, because the cruise and the tour come first in every option.

The release day runs as follows: checkout and timer load at T−30 h, separation at about T−5.2 h, the 328 m/s burn at T−4.4 h, despin and a 67° slew, coast with telemetry, core preheat in the last hour, impact 12 min after the flagship's closest approach, and 30 min of post-impact relay. The flagship then returns the stored data to Earth (Section 6.4).

6.4 Communications, Data Use and Relay

SMAD sizes a data system from what each sensor produces and what the link can carry [3, pp. 544–545]. Table 7 lists PERCI's products in the order they are sent, with their use, and Figure 15 compares them with what the UHF relay returns.

The geometry sets the data volume. The relay is a single pass: the flagship is 5.4–6.5 Mm away at impact and is leaving, so the post-impact return is 0.15 MB at the median (0.04–0.55 MB, 5–95%). This is 16%



Figure 14: Mission timeline from launch to impact (a) and the release-day sequence (b). The penetrator is stored for 16.0 years before use, not the 13.4-year cruise that the 2025 report used for its cold-soak risk. Dates from the UOP study [1]; the release-day times follow Sections 5.1 and 6.2.

of the 7.6 Mbit of payload telemetry planned for a two-week Ganymede penetrator [40], and it is delivered in 30 min rather than over weeks of orbiter passes, as in Lunar-A and MoonLITE (30 kbit/day) [35, 38]. The accelerometer window (Q5.1c), the temperature decay (Q8.2f) and the housekeeping fit; the seismometer record does not, at the median. On the flagship side, the whole flight volume of 15.8 MB (126 Mbit) is small. UOP returns it in 1.8 h at its 19.5 kbps Ka-band maximum, or 5.0–5.9 h at X-band, which is about 1.0% of the 13–15 Gbit returned per 34-day orbit [4]. Store-and-forward on the flagship is therefore not a constraint; the penetrator-to-flagship pass is.

Second-pass relay, considered and rejected. The second Umbriel flyby comes 103 days later. Surviving to it would need 1.2–2.5 kWh for a 0.5–1 W hibernation load (ASSUMED), that is 26–53 LSH20 cells (2.6–5.3 kg), which is the scale of Philae's 1.5 kWh primary battery [56]. The heat leak is the real obstacle: at the v2 core-to-shell conductance of 0.08–0.25 W/K, holding −40°C in 70 K ice costs 13–41 W continuously. One watt would need $G \leq 0.006$ W/K, about 40 times better isolation, or radioisotope heater units. The visibility of the site on the second flyby is also unknown. We therefore keep the single-pass design.

6.5 Reliability and Cold Soak

SMAD's reliability process starts by keeping the design simple and then removing single-point failures [3, pp. 764–765]. PERCI v2 is single-string throughout. Table 8 and Figure 16 list the chain in flight order, with the evidence available for each link.

Two of the nine links have a model value and one has flight heritage. The other six have no datum at all, so a product of link reliabilities would be invented, and we do not compute one. A reference class is more honest. Treating outcomes as Bernoulli trials with a Jeffreys prior, the only penetrators that reached their target (DS-2, 0 of 2) give a posterior mean of 0.17 (90% upper bound 0.57). The deep-space secondaries released from a carrier (Huygens, the Galileo probe, MINERVA-II1A and B, MASCOT, SCI, DCAM3 and LICIACube succeeded; MINERVA and both DS-2 probes failed; Philae is counted as a failure here) give 0.65 (0.43–0.85). These are base rates for the class, not predictions for PERCI. They show that released secondaries usually work and that penetrators have no successful precedent. Heritage practice also shows where redundancy is cheap: Huygens used a triply redundant mission timer, three pyronuts and two S-band chains [47].

Cold soak. Over 16.0 years the Li-$SOCl_2$ cells lose at most 15% of their capacity at +20°C (self-discharge

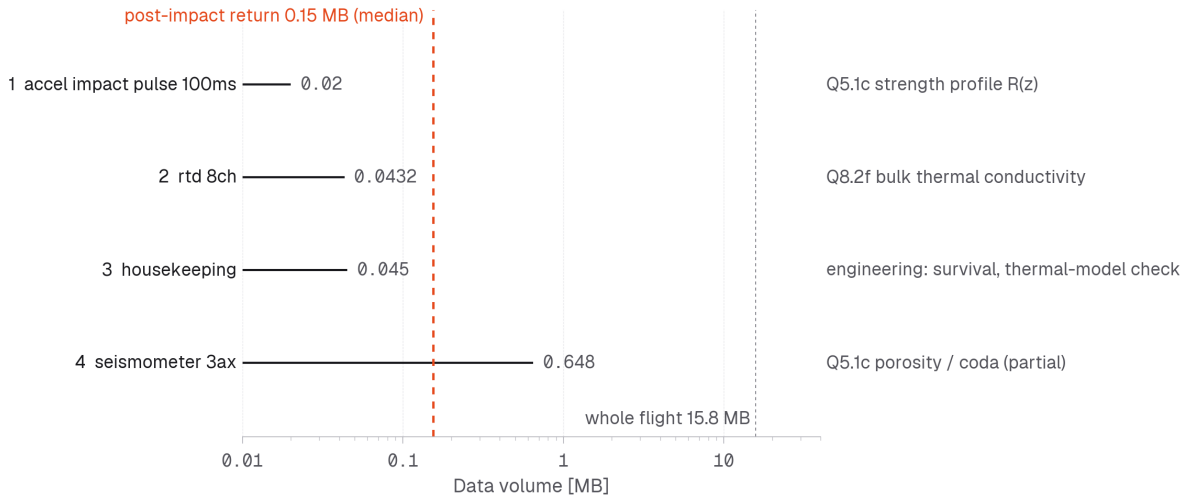


Figure 15: Data products in priority order, the volume sent and what each is used for. Only the first three fit the median post-impact return of 0.15 MB. Link Monte Carlo of Section 5.5.

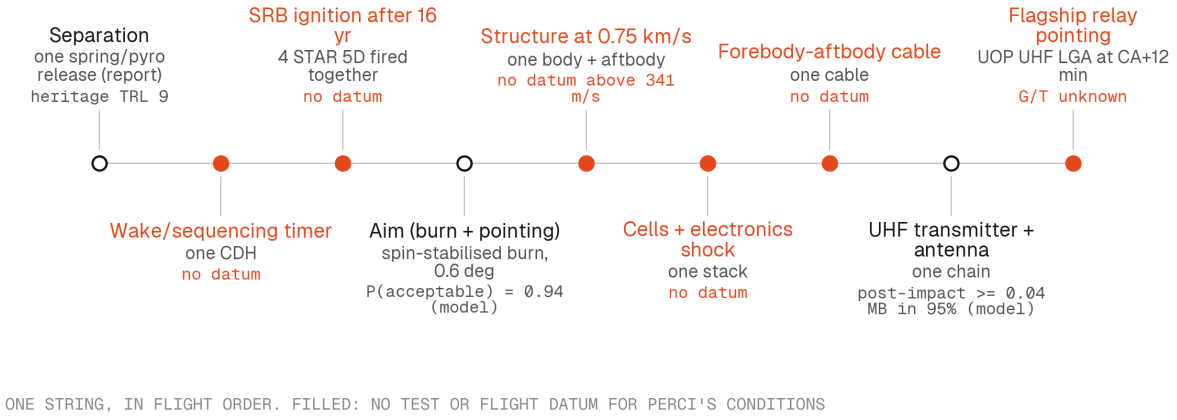


Figure 16: Single-string chain from separation to relay. Six of the nine links have no test or flight datum at PERCI's conditions. Model values from Sections 5.1 and 5.5; heritage from Table 6 and [33, 47].

below 1%/yr [57]), so at least 85% remains, and less is lost when cold. The passivation after long storage is the larger effect: the Huygens batteries had to be depassivated after seven years of inactivity [58], whereas Philae's LSH20 cells worked after a ten-year cruise [56]. The motors are the open item. We found no STAR 5D storage limits; analogous Northrop Grumman motors are stored at +20 to +100°F (Orion series) or +30 to +100°F (GEM 40) [59], that is −6.7°C or −1.1°C at the lowest. Figure 17 shows the heater power needed to hold −6.7°C in deep space with no view to the warm flagship.

The heater needed is 1.4–13 W, against the 2 W allowed from the orbiter and the 0.64 W v2 cruise load, which has no motor heater in it. Only MLI as good as the report's best case ($\epsilon^* = 0.005$) on a 1 m^2 envelope fits. In practice PERCI must be mounted inside the flagship's thermal enclosure, or the 2 W allowance must grow. This is a new requirement on UOP.

6.6 Cost Class

The 2025 report adopted the SIMPLEx guidelines (Section 4). To test that class, we apply the small satellite cost model of SMAD Table 20-6 [3, pp. 796–797]. The bus cost is $781 + 26.1\,X^{1.261}$ (FY2000 k$, X the bus dry mass, valid for 20–400 kg), the payload is 0.4 times the bus, and integration and test, program level, ground support and launch support add 0.139, 0.229, 0.066 and 0.061 of the spacecraft cost. The bus dry mass is

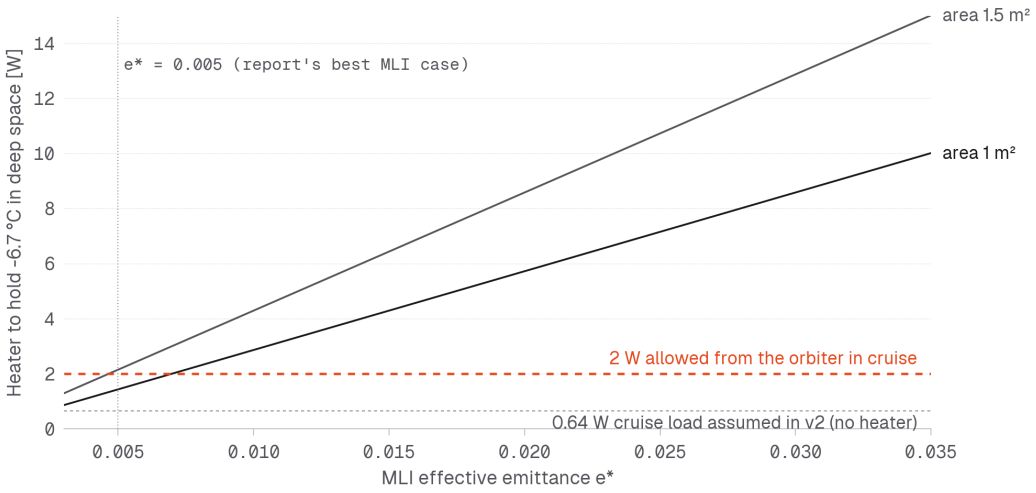


Figure 17: Heater power needed to hold the motors at an analogous storage floor of −6.7°C in deep space versus MLI effective emittance, for envelope areas of 1.0 and 1.5 m^2. Only $e^*A \leq 0.0070$ m^2 fits the 2 W cruise allowance. Areas and emittances are ASSUMED; sunlight at 19 AU is neglected.

123 kg (CBE) to 152 kg (MEV) once the 9.6–11.2 kg payload is removed. Escalated with the consumer price index (a proxy; [60]), the model gives \$46–59 M in 2024 dollars, with a standard error of about \$7 M on the bus term alone. Figure 18 compares this with class caps and heritage programs.

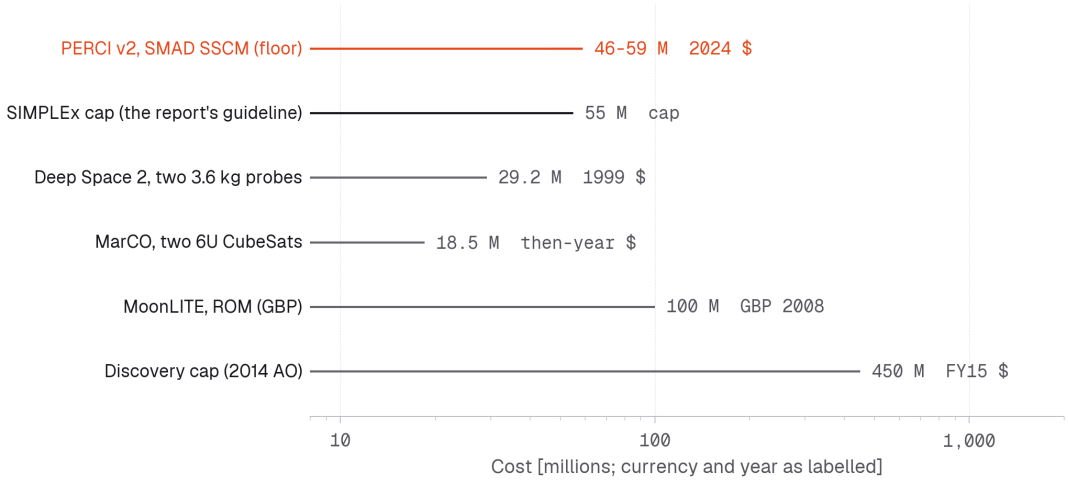


Figure 18: PERCI v2 cost from the SMAD small-satellite cost model against class caps and heritage programs. The model gives a floor of \$46–59 M, already at the SIMPLEx cap. Currencies and years differ as labelled and are not converted. Sources: [3, 34, 38, 61–63].

The model is fitted to Earth-orbiting small satellites, assumes a protoflight approach [3, p. 798], and sees neither a 20,000 g qualification program nor a 16-year cruise, so its value is a floor. For scale, DS-2 cost \$29.2 M (1999 dollars) for two 3.6 kg probes [34] and MoonLITE's rough estimate was £100 M for four penetrators and an orbiter [38]. As a result, the SIMPLEx framing (a \$55 M cap [61]) is not supported for a 150–190 kg penetrator. The floor is 1.1–1.4% of the \$4.2 B UOP estimate [64]. A more realistic route is a contributed element, as Huygens (ESA), MASCOT (DLR with CNES) and LICIACube (ASI) were [44, 47, 52].

6.7 Planetary Protection

The 2026 COSPAR policy treats bodies whose outer layer is mostly water ice and that are nearly round as Icy Worlds. Landers and probes default to Category IV, and Category II is allowed only if analysis shows a probability below 10^{-4} that any part of the spacecraft reaches an environment above −28°C within 1000 years

[65]. Titania is on the policy's list of Icy Worlds; Umbriel is not on the list we read, although it meets the definition, so we treat it as one. Two features of PERCI make a Category II argument harder than for an orbiter. First, the impact releases 33.5–42.0 MJ. Warming 70 K ice to the melting point and melting it takes 613 kJ/kg (heat capacity from [24], latent heat 333.5 kJ/kg), so the impact could melt at most 55–69 kg of ice if all of its energy went into the ice (an upper bound). Second, the warm core of Section 5.3 is itself a buried environment near or above the limit: in 62% of the sampled start temperatures it begins above −28°C. The JUICE precedent rests on a transport analysis that showed at least 7,000 years for Ganymede [65]. PERCI needs the same analysis for Umbriel's regolith and ice, or it must plan for Category IV bioburden controls on the forebody. Neither cost is in the 2025 report or in our cost floor.

6.8 Science Value per Kilogram and a Small-Penetrator Alternative

The results so far raise the question whether a penetrator of this size is the right way to put measurements on Umbriel. To answer it, we compare four options against the same traceability rows (Table 1), scoring 1 for an achievable row and 0.5 for a partial one (Table 9). Mass is converted to a launch-mass equivalent with the UOP launch-to-dry ratio of 7,235/2,756 = 2.63 [66]. Each kilogram carried through the mission costs about 2.6 kg at launch; this is an upper bound for PERCI, which is released after orbit insertion.

Figure 19 shows the trade. A flagship-only campaign scores 0 in situ: remote thermal inertia samples only the top centimetres, and no remote method gives a strength profile. PERCI v2 scores 3.0 for 420–500 kg of launch-mass equivalent. A 13 kg-class penetrator scores 2.5 for 55–81 kg, about one sixth of the mass, and loses only the descent imagery beyond a 10 g imager of the MoonLITE kind [38]. A MASCOT-class soft lander would need about 1.19 km/s of braking and terminal guidance with no heritage at this approach speed, and it scores 2.0 because it gives no strength profile.

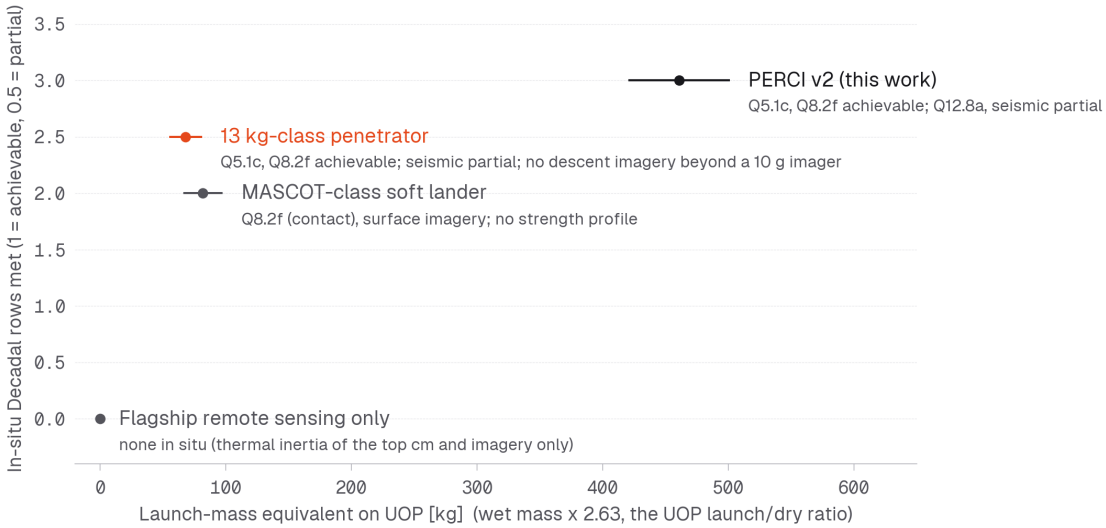


Figure 19: In-situ science rows met versus launch-mass equivalent for four ways to put measurements on Umbriel. A 13 kg-class penetrator meets 2.5 of PERCI's 3.0 rows for about one sixth of the launch-mass equivalent. Scores from Table 1; masses as in Table 9.

The small body only helps if it survives, so we ran it through the penetration model of Section 5.2 with the same Sobol-sampled targets (1,536 samples, 11–15 kg, radius 0.04 m, 0.5 m long, 750 m/s). Figure 20 compares it with the v2 body.

The sectional density of the small body is 4.4 times lower. In hard ice it stops at 0.7–3.2 m (5–95%) rather than 2.2–11.7 m, in regolith over ice at 2.9–15.3 m and in porous regolith at 31–139 m, against 120–600 m. The shallower burial shortens the aftbody cable and cuts the burial loss of the link. The price is the shock: 10–60 thousand g in hard ice, with half of the cases below the 24,000 g that the ESA test survived and 73% below the 35,000 g of the 2016 gas-gun test [13, 45]. This demonstrates that a 13 kg-class penetrator is not ruled out at Umbriel and would need about one sixth of the mass, but it moves the risk from burial to shock. It is a Phase-A alternative, not a result.

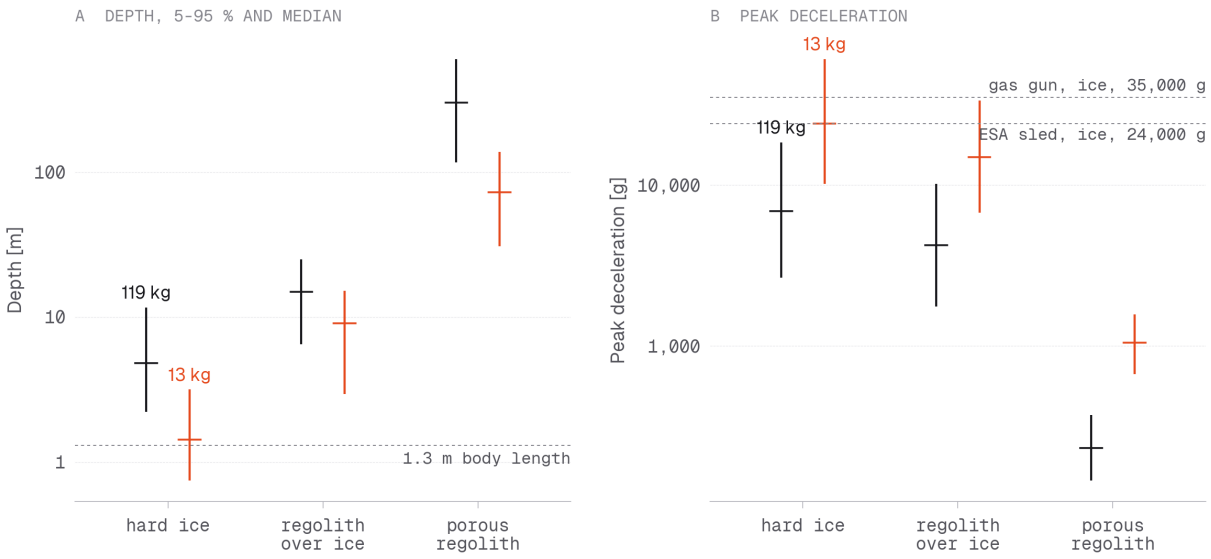


Figure 20: Depth (a) and peak deceleration (b) at 750 m/s for the v2 body (119 kg, 0.058 m radius) and a 13 kg, 0.04 m body over the same sampled targets. The small body stops 3–4 times shallower at about 3.5 times the deceleration; its hard-ice median equals the 24,000 g ESA sled test, but 27% of hard-ice cases exceed 35,000 g. Reference levels from [13, 45].

7 Discussion

This section discusses what the results mean for the risk posture and the design, how the remaining uncertainty would be retired by test, and where the analysis itself could be wrong.

7.1 Risk Register (FMEA Recomputed)

The risk priority number is computed as RPN = $S\times$(P score $\times$ D score), following the report's convention (C.3). The P scores are binned from the model probabilities (bins < 1, 1–5, 5–20, 20–50, $> 50\%$). The report's cost, vendor and instrument-activation risks (CST-03 80, SHD-01 75, SCI-06 60) lie outside the models and are carried unchanged. Figure 21 compares the RPNs of the report concept and v2, and Table 10 gives the recomputed register.

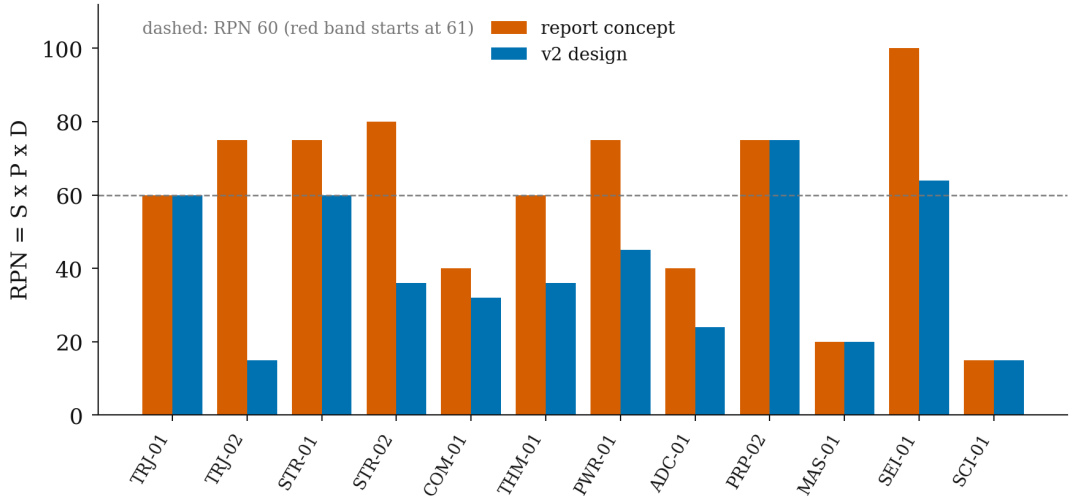


Figure 21: RPN of the report concept versus v2.

The highest residual risks after v2 are SRB ignition after 16.0 years of storage (RPN 75, with no verified datum; the report counted 13.4 years), seismometer shock survival (64), peak deceleration $> 10{,}000$ g (60), and impact acceptability (60, dominated by pointing and burn repeatability).

7.2 v2 Design Summary

Table 11 lists every value changed from the report, with the reason and the source of each change.

Growth option. The Phase-A study should evaluate the 12–24 h release with a monopropellant or small SRB stage (a propulsion mass of 3–9 kg instead of 26 kg, and 5–8× the descent data), *conditional on the actual UOP tour state*, which fixes the flyby orientation. This option removes the single largest non-robust assumption (the orientation) and the largest mass line.

7.3 Verification Plan

SMAD's qualification flow tests a design on a dedicated or protoflight unit with functional tests between each environmental exposure [3, pp. 520–525]. The DS-2 board added the lesson that matters most here: a complete, electrically powered system was never impact-tested, the flight battery lot was never impact-tested, and the antenna was never tested in the Mars environment; it recommended "test as you fly" and at least one powered, flight-quality probe through system-level qualification [33, Secs. 3.4.2, R18–R19]. The following tests and inputs would close the open items identified above:

1. **High-speed penetration test** (gas gun or rocket sled) of a 1/4-scale ogive penetrator into 85–100 K cold ice and ice-regolith layers at 0.5–0.9 km/s, measuring $R(z)$, depth and the g-history. This closes TRL 2 $\rightarrow$ 4.
2. **Shock qualification** of the cell stack, seismometer, accelerometer chain and insulation to 10,000 g for 1 ms, with DS-2-type cell hardware for impact (80,000 g, [22]).
3. **Cold cell characterisation**: capacity and pulse resistance of candidate Li-$SOCl_2$ cells at 233–253 K and 0.1–1 A, after accelerated 16-year storage (voltage delay and depassivation, as Huygens needed after seven years [58]).
4. **Thermal-vacuum** test of the warm core in a cryogenic ice/regolith simulant block, to measure G_{ins} and the heater duty directly.
5. **Link**: the UOP UHF LGA gain pattern and receiver noise temperature (a requirement request to the UOP project), and an antenna range test on the penetrator body with an ice-simulant burial.
6. **Trajectory**: the UOP tour ephemeris for the 2047 Umbriel encounter and the navigation covariance, followed by a Monte Carlo with finite-burn and spin dynamics (6-DOF).
7. **SRB**: a STAR 5D cold-soak and ignition test after thermal cycling (operating limit -55°C [23]), and the manufacturer's storage limits, which set the cruise heater of Section 6.5.
8. **Replica-scaled high-speed tests.** At the same impact speed, a geometric replica at 1/4 scale keeps the stresses, has 1/64 of the mass (119 to 1.9 kg) and sees 4 times the deceleration, because the force scales with a^2 and the mass with a^3. Small-calibre guns can therefore reach 0.75 km/s with representative targets, while the full-scale sled (Pendine, 300–341 m/s [13, 39]) is kept for the system-level test.
9. **Powered system-level impact test** of a flight-lot unit with cells, electronics, cable, aftbody and antenna operating, followed by a post-impact link test through an ice-simulant burial (DS-2 R18–R19 [33]).
10. **Separation and checkout**: a spin/eject device test in the Huygens manner [47], an umbilical checkout after integration, and a go/no-go rule for the T$-$30 h timer load (the Philae and MINERVA lessons, Section 6.2).
11. **Planetary protection**: an Umbriel transport analysis to the COSPAR Category II standard, or a Category IV bioburden plan for the forebody (Section 6.7).

7.4 Sources of Error and Model Limitations

Several simplifying assumptions in this analysis warrant discussion regarding their effect on the results.

Planar delivery model: The delivery model is planar, and the flagship's out-of-plane motion is handled only by a first-order variational equation.

Flyby orientation: The flyby orientation is unknown. The four-case spread is a bounding assumption, not a covariance.

Error budget: The error budget is ASSUMED; there are no heritage numbers for the burn repeatability, the thrust-vector offset or the Uranian-satellite ephemeris error in 2047.

Penetration model: The penetration model is a cavity-expansion model with assumed material inputs. Ice strength at 85 K and high rate has no verified datum, and no experiment exceeds 341 m/s.

Thermal model: The thermal model is radial 1-D with an end-factor. It has no impact-heated sheath, and the insulation properties are UNVERIFIED.

Link budget: The link results depend on the assumed antenna gains and noise temperature.

Seismic model: The seismic model is an energy-flux estimate, good to a factor of 3–10.

Reference-class reliability: The base rates of Section 6.5 come from 2 penetrators and 12 released secondaries with different designs and environments. They are not predictions for PERCI, and a different choice of members (for example counting Philae as a success, which gives 0.73) moves them by about 0.1. No conclusion rests on their exact values.

Cost model: The SMAD small-satellite cost model is fitted to Earth-orbiting spacecraft of the 1990s and is escalated here with the consumer price index rather than a NASA index. Its standard error alone is about $7 M on the bus, and it omits high-g qualification and a long cruise. The direction of the error is therefore upward, which strengthens the conclusion that the SIMPLEx class does not fit.

Cruise thermal estimate: The motor heater assumes radiation to deep space through MLI with no view to the flagship and no sunlight, and uses an analogous motor's storage floor. Mounting inside the flagship's enclosure could remove most of the load. The estimate sets a requirement to confirm, not a design.

Options trade: The science scores count traceability rows equally, the small-penetrator and lander masses carry ASSUMED subsystem allocations (±25%), and the launch-mass factor of 2.63 is an upper bound. The ordering of the 13 kg penetrator against PERCI v2 holds over this range, because the mass ratio is about six.

Target-class weights: The prior weights of the target classes are ASSUMED, which is why the results are given per class.

Table 12 gives our own assessment of the technology readiness of each element of the concept.

7.5 What a Reviewer Would Challenge

The objections a reviewer would most likely raise, and our responses, are as follows.

- *"Your minimum-ΔV solutions are optimal for one geometry."* Yes. The spread over four orientations is our response: the 4.4 h design is robust (P 0.94 in all four), while the early-release designs are not.
- *"Cavity-expansion models are not valid at 0.75 km/s in ice."* Agreed; no model is. The distributions bound the uncertainty, no design relies on a single class, and verification step 1 is the closing action.
- *"The thermal result depends on insulation you have not shown to survive 5,000 g."* Agreed (TRL 3). However, the unheated result is also valid at the report's own $G = 0.92$ with an 11 kJ/K core.
- *"The link numbers are pessimistic."* They are assumption-driven, and the required inputs are the UOP UHF G/T and the penetrator antenna design.
- *"Why not fly a 13 kg penetrator instead?"* It may be the better design. It meets 2.5 of 3.0 in-situ rows for about one sixth of the mass, but its hard-ice shock reaches 24,000 g at the median and exceeds 35,000 g in 27% of cases, so it trades burial risk for shock risk (Section 6.8).
- *"Descoping Q5.1a removes the point of a seismometer."* A single co-located station cannot constrain the interior structure in 30 min (the direct range is 17–316 km against a 585 km radius), and we say so.

8 Conclusion and Recommendations

This work audited the 2025 PERCI concept study, rebuilt its delivery, penetration, thermal, power and link models as deterministic, tested codes with stated uncertainty, and added the mission-level elements and the comparison with flown practice that the study lacked. The concept is not invalidated, but its headline numbers are not defensible as published.

At the subsystem level, the results show that aiming for the minimum-ΔV central hit lowers both the retarget ΔV and the impact speed, and that it also removes the need for the 180° flip. The large heater power was found to be a consequence of the thermal model rather than of the environment, and a warm, insulated core removes it for the 30-minute requirement. In contrast, the link does not close as written and only a small fraction of the planned data can be returned, so the UOP UHF receive performance becomes a required input rather than an assumption.

At the mission level, the v2 baseline meets the delivery, thermal and energy requirements with a descoped science set, while the mass fits only under the dry interpretation of the limit. Against heritage it is far outside the tested envelope in both speed and mass, its single relay pass returns a small post-impact data set, its motors need heat in cruise that is not budgeted, its cost is at the SIMPLEx cap before qualification, and its planetary-protection category is open. A much smaller penetrator meets most of the same science for a fraction of the mass.

In general, the results demonstrate that a flyby-released penetrator for Umbriel is not ruled out as a secondary payload, provided its claims are stated at the strength the models support. The dominant open risk is the penetration physics at high speed into cold ice, for which no model is validated. We recommend the following:

1. Carry a 13 kg-class penetrator as the Phase-A baseline alternative to the 119 kg body, and decide between them on shock tests, not on analysis.
2. Begin with replica-scaled gas-gun tests at 0.5–0.9 km/s into 85–100 K ice and layered regolith, followed by one powered, flight-lot, system-level impact test.
3. Release with a flagship-side spin/eject device, separate about 50 min before the burn, and require an umbilical checkout and a go/no-go rule before the timer load.
4. Ask the UOP project for the UHF G/T, the tour state of the 2047-equivalent Umbriel encounter, a pointing allocation for 30 min after closest approach, and either a mount inside the thermal enclosure or a cruise heater allowance above 2 W.
5. Treat PERCI as a contributed element above the SIMPLEx class, and fund the Umbriel planetary-protection transport analysis at the start of Phase A.

Future work should repeat the delivery and link analyses once the UOP tour state and UHF G/T are known, and should replace the reference-class reliability with test-based estimates as the qualification program proceeds.

Acknowledgments

The author thanks the 14-person 2025 PERCI study team, whose concept study is the basis of this re-analysis; the thermal, impact, trade-study and CAD work in that study was the author's contribution. The course instructors and mentors are also thanked for their guidance during the original study.

A Audit Findings

Table 13: Audit findings (A: lead, impact/thermal/trajectory; B: independent, comms/power/ΔV/ADCS/mass/risk). Sev.: Crit. = invalidates a headline conclusion, High = changes a design number materially. Verdict: ERR confirmed error, UNS unsupported, ASM assumption, OK report is right.

ID	Sev.	PDF p.	Claim as printed	Independent recomputation	Verd.
A-01	Crit.	p.64 Table 34, p.10, p.17	Thermal: 140 W heater for 30 min	Table 34 is reproduced (rms 3.4 W) by a 1-node battery-only model with G=0.92 W/K and C=6.2 kJ/K. A realistic insulated core (11 kJ/K) at the same G needs 0 W for 30 min (12.6 W from a -39 C start). 140 W follows from the 1 W/K mount and battery-only thermal mass assumptions	ERR
A-02	High	p.55-57, Table 29	1-node analysis: 7.9 W best case	The model is a 1.6x1.06x0.96 m aluminium box (SIMPLEx envelope), not the 0.116 m-diameter penetrator (0.072 m3); the 70-node model is also a cuboid	ERR
A-03	High	p.51	Target resistance R_t = 0.6 GPa (eq B.5.3.4)	eq B.5.3.4 with sigma=0.15 GPa, E=9.1 GPa gives 0.454 GPa; 10.4 MN is reproduced only with 0.6 GPa (8.8 MN with the formula)	ERR
A-04	High	p.49-51	Ice flow stress 0.15 GPa from Glen's law	Glen's law is a creep law (below ~1 MPa); inversion at 0.15 GPa implies a strain rate of 4.4 /s while the text calls the ratio '4' a strain-rate ratio; A=1.3e-24 is for temperate glacier ice, not 70 K ice	ERR
A-05	High	p.50-51	Tate-Alekseevski blunt-face force for an ogive CRH-5 nose	Inertial coefficient N=0.5 (blunt) vs 0.065 (ogive CRH5): 7.7x over-prediction of the dynamic term; the report's second model (5.3 MN) is reproduced	ERR
A-06	High	p.15-16, 48-52	Impact: 'range' 5.4-10.4 MN, 3.4-8.3 m	Sobol-sampled models (3 target classes): hard ice depth 2.2-11.7 m at 2.6-18 kg; regolith over ice 6-24 m; porous regolith 120-600 m at 140-360 g. The report range is the stiff-ice corner, not an uncertainty band. The empirical Sandia/Young equation (warm ice) gives 39 m at 900 m/s	UNS

ID	Sev.	PDF p.	Claim as printed	Independent recomputation	Verd.
A-07	High	p.15-16, 51	No validation of any impact model at 0.75-0.9 km/s	Highest documented full-scale ice impact: 341 m/s, 24,000 g, 20 kg (S23); Young equations valid below 1219 m/s with no soft-material data at very high speed (S27). No flown penetrator has operated (S14)	ASM
A-08	Med.	p.16 vs p.52	Summary vs appendix impact numbers	6.5-16 MN and 2.7-3.5 m (Sect 6.6) vs 5.4-10.4 MN and 3.4-8.3 m (B.5.3)	ERR
A-09	High	p.15	Aft flare 'prevents overpenetration'	Flare radius 0.12-0.2 m raises hard-ice peak g from 4.2 kg to 16-42 kg and still leaves nose depth 2-3 m (more than the 1.3 m body): antenna buried. DS-2/Mars-96 use a split forebody/aftbody with a surface antenna (S10)	ERR
A-10	High	p.19 Table 2	Umbriel: 415 m/s → 901 m/s, 86 deg, 4.2 h	Independent central-hit model: 401 m/s → 862 m/s, 3.76 h (within 4% / 10%). Reproduced	OK
A-11	High	p.19 Table 2, p.18	dV 415 m/s needed; 450 m/s requirement	Minimum-dV central hit at 4.4 h is 323-328 m/s over the four unknown flyby orientations (v_imp 0.75 km/s): 22% lower dV and 13% lower impact speed than the report row	ERR
A-12	Med.	p.19 Table 2	Ariel, Oberon, Titania rows (479/414/820 m/s)	Not reproduced: Ariel 257 m/s → 0.94 km/s, Oberon 117 → 1.71 km/s, Titania(2045) 75 → 3.0 km/s. The objective used for those rows is not stated (they appear to trade dV for lower speed)	UNS
A-13	High	p.8, p.26 (SYS_SD_6)	'Titania unreachable' (820 m/s)	With the 2047-11-07 Titania flyby (0.47 km/s, 6,505 km) a central hit needs 377 m/s at 4.4 h (v_imp 0.86 km/s); the report row used the 2045 fast flyby	ERR
A-14	High	p.70-72	Retrograde burn then 180 deg flip	The minimum-dV burn direction is 67 deg from the impact velocity vector (4 orientations, 4.4 h): a slow ~67 deg slew replaces the 10 deg/s flip (see B-28)	ERR
A-15	Med.	p.14	'Crary waves' for interior structure (Q5.1a); coda from one 30-min station	Crary waves need a floating ice shell over an ocean (S38); no ocean established on Umbriel (S07). One co-located station gives near-surface Vp/attenuation only	UNS
A-16	Med.	p.12	Q4.4b composition via the flagship's 'Lucy L'Ralph' spectrometer	L'Ralph is a Lucy payload; presence on UOP UNVERIFIED (UOP payload list not confirmed); plume observation geometry at 5.4-6.5 Mm not shown	UNS
A-17	Low	p.34	Accelerometer 350B41 (-23..66 C)	An AC-coupled channel with a 4 Hz corner biases velocity from a 10 m pulse by 27%; minimum operating temperature -23 C vs core -40 C	ASM
A-18	Med.	p.57-60	70-node model: MLI e*=0.01 buried in ice	The vacuum-radiative MLI concept is not valid in compressed contact with ice; ice cases use a constant heat load, not a thermostat; no conduction into ice from first principles	ASM
B-01	Crit.	p.40-41, 10, 7	Link closes at 2 GHz to a 1 m orbiter antenna (Table 1 'S-band'; Sect 3.1.1 'UHF')	UOP lists X, Ka and a UHF LGA, no S-band. UHF 401.6 MHz at 6.5 Mm with assumed gains: 6 kbps +4..+2 dB (spread -6..+12), 20.1 kbps -1..-3 dB	ERR
B-02	Crit.	p.10, 15	'15 dB at 10 kbps, 1 dB at 2.14 Mbps' at ~6,500 km	Table 19 as printed: 14.3 dB (1.1 kbps), 4.8 dB (10 kbps), 1.8 dB (20.1 kbps), -18.5 dB (2.14 Mbps). The claims use EIRPs 8 dB apart; 2.14 Mbps is 106x the 20.1 kbps cap	ERR
B-03	High	p.40	RF power 5 W, EIRP 5.67 dBW	RF = 3 W (DC 5 W x 0.6); EIRP 11.4 dBW (13.66 at 5 W)	ERR
B-04	High	p.41	Rx gain 15.0 dBi (1 m, 0.6, 2 GHz)	24.2 dBi (15.0 is a 0.35 m dish)	ERR
B-05	High	p.42	B = 0.125 MHz, C/N 80.2 dB, Rmax 3.3 bps, 'Exceeds Shannon? NO', 'Above sensitivity? NO'	C/N = 20.2 dB; 80.2 dB corresponds to B = 0.125 Hz (Hz/MHz slip); Rmax = 842 kbps; the flags contradict the +30.6 dB margin	ERR
B-06	High	p.41	Ts = 70 K 'ground station at room temperature'	The receiver is on the orbiter; 290-500 K costs 6-8.5 dB	UNS

ID	Sev.	PDF p.	Claim as printed	Independent recomputation	Verd.
B-07	Low	p.40-41	Prx -139.1 dBW	-137.9 dBW: line and pointing losses subtracted twice (1.18 dB)	ERR
B-08	Med.	p.41-42	LoRa SF7 'modulation rate 976' with 8PSK Eb/N0 8 dB	976 is a symbol rate; LoRa is spread-spectrum so the 8PSK requirement does not apply; 1 kbps = 1024 bps	UNS
B-09	Med.	p.10	'-104 dBm at 10 deg mismatch, -118 dBm at 45 deg'	Not reproducible: the patch pointing-loss spread is 4.7 dB, not 14 dB	UNS
B-10	High	p.10	Memory 50 GB, 20 GB needed, 2.14 Mbps	The 20.1 kbps cap over 4.2 h + 30 min returns at most 42.5 MB; 20 GB would take 92 days	ERR
B-11	High	p.15, 18	In line of sight after impact because flight (4.2 h) < time before CA (4.4 h)	Closest approach is 12.5 min after impact; slant range 5.4-6.4 Mm; flagship above horizon in only 75% of impact solutions (70% at +30 min)	UNS
B-12	High	p.15-17, 41, 50-53	Flare keeps antenna at surface; 0.05-0.5 dB/m at 2 GHz	Depth exceeds body length (1.3 m); burial loss 0.1-4 dB at 2 GHz needs tan d 1.6e-4..1.6e-3 (no measurement); patch half-angle 35 deg vs flagship zenith 0-149 deg	UNS
B-13	OK	p.68	Table 37 sums and 705,145.5 J	29.175 W and 143.76 W reproduce; 705,145.5 J = 29.175 W x 4.25 h + 143.76 W x 0.5 h; 0.5936 kg and 0.7898 L reproduce	OK
B-14	Med.	p.10	'37 W avg / 140 W pulse'	37 W not reproducible (29.175 W descent, 41.2 W time-average); 140 W is a 100% duty 30 min load	ERR
B-15	Med.	p.8, 24, 11, 19, 68	Pre-impact time 4.2/4.25/4.4/4.5/4.6 h	Five different values; deploy to end of 30 min = 5.1 h, more than the 4.5 h survival requirement and the 5 h wheel life	ERR
B-16	Low	p.68, 17	330 Wh/kg and 248 Wh/L	Conservative vs LS33600 680 Wh/kg; mutually inconsistent for one 21 g, 21.6 cm3 cell; stack holds 1,331 Wh = 6.8x the need	ASM
B-17	Med.	p.17-18	4 x 2.75-3.5 V = '12 V'; 4s48p	11-14 V; capacity at -40 C 16-31% at 0.1-1.3 A; usable 224-691 Wh vs 195.9 Wh; voltage delay not analysed	ASM
B-18	Med.	p.68, 16	Heater 140 W x 0.07 duty = 9.8 W in descent	9.8x the '<1 W' survival heater; no thermal case supports it	UNS
B-19	Med.	p.13-14, 34, 68	OSIRIS-REx cameras (4 kg, 10 W each) vs Table 37 'Simera Sense' 5 W; Artius 0.6 W vs MEMS 1.5 W	Instrument table and budget disagree; two Table-13 cameras are 40 L vs a 6.8 L tube	ERR
B-20	Low	p.46, 68	ADCS 60 W x 0.035 = 2.1 W	Arithmetic OK; 60 W is per wheel; 0.035 x 4.25 h = 8.9 min vs '5 minutes'; star tracker/IMU absent	UNS
B-21	Med.	p.79 Table 46	Cell trade: DS-2 LTC wins 3.746 vs 3.6725	Totals reproduce, but a blank voltage-stability cell is scored 0 (4.444 with 0.775); margin 0.074 is smaller than the impact row 0.645; 5,000 g must-have is below the report's own 9,600 g; DS-2 cell is custom (COTS requirement); both DS-2 probes were lost	ERR
B-22	Low	p.71 Table 40	476 m/s with 4 motors	The series 139/265/379 matches base 120 kg exactly; n=4 gives 483.1 (base 119 kg: 483.3 with STAR 5D masses)	ERR
B-23	High	p.19, 71	dV margin 476 vs 450 m/s	Holds only with zero growth: 9% base-mass growth removes it; 25% growth gives 401 m/s	ASM
B-24	High	p.8, 26, 19	Umbriel and Titania required; Table 2 'lowest flyby velocity'	Titania row (820 m/s) unreachable with 4 SRBs, uses the 2045 fast flyby while Fig 14 lists a 0.47 km/s flyby	ERR
B-25	Low	p.70	STAR 5D data	Matches the NG catalog (10.23 kg, 3.23 empty, Isp 256 s); omitted: burn 3.03 s, 5,565 N, 13.6-19.4 g	OK
B-26	Med.	p.19, 70	SRB after 13.4 yr cruise	STAR 5D operating limit -55 C (storage -62 C); no SRB thermal case; 17 deg cant loss (4.4%) not mentioned	UNS
B-27	Med.	p.51	Impact mass 110 kg = 150-40	The budget gives 159-40 = 119 kg (149 kg at MEV)	ERR
B-28	Crit.	p.45-46, 71-72	10 deg/s SRB-ejection flip stopped by 60 mNms wheels	Needs 2.6-4.0 N m s: 43-66x the wheel; '30 mNms 5 minutes' mixes momentum and time; 1.5% impulse matching required	ERR
B-29	Low	p.71-72	Spring ejection of SRB pairs	0.5-1.2 m/s relative speed; lateral recoil 2.6-6.3 cm/s = 0.4-1.0 km miss over 4.2 h, not budgeted	ASM

ID	Sev.	PDF p.	Claim as printed	Independent recomputation	Verd.
B-30	Med.	p.16, 46	Spin-up 'tiny solid rocket motors'	Spin of 860-1,220 rpm needed for 1 deg with 1 mm/0.5 deg SRB offset; spin-up impulse 0.26-0.82 kN s (not tiny); Iz/It ~ 0.06 is a minor-axis spinner, unstable under dissipation	UNS
B-31	Low	p.45	Gravity gradient 1.5e-9 N m	Right order for Titania/Uranus; Umbriel's own gradient is 1.4e-8 to 1.3e-5 N m at 5,939-600 km (24 mNms integral = 40% of the wheel)	ERR
B-32	Low	p.46	Table 22 caption/units	'60mMms' (momentum unit in a torque column), informal caption, wheel life 5 h vs 5.1 h	ERR
B-33	High	p.9, 24	159 kg 'wet' of 180 kg; '12% margin'	The limit is stated dry (A.3); 25% propulsion = 4 x 10 kg so 159 is wet (dry ~131). S-120 growth of 26% predicts 193-195 kg wet (over by 13-15 kg); 21 kg = 11.7% of the limit, 13.2% of CBE	ERR
B-34	Low	p.9-10, 24, 55	Volume 0.072 m3, '>80% unused'	Consistent with 4 x 9,250 cm3 = 51.6%; the limit is quoted 0.42, 0.46, 1.19 m3 and 1.63 m3 in the thermal model	ERR
B-35	Med.	p.17-18, 34, 68	Battery 4.15 L in a 5.3 L body tube	78% of the interior; 1.4x packing exceeds it; a 20 L camera does not fit	ASM
B-36	Low	p.9	Mass budget as percentages only	14.3 kg unallocated (itemised to a 0.45 kg residual in power.py)	UNS
B-37	OK	p.80-81	RPN = S x band score	CST-03 80 = 5x16, SHD-01 75 = 5x15, SCI-06/PRP-01 60 = 5x12 are consistent	OK
B-38	Low	p.10, 80	'56 risks, 14 top risks, top three'	55 IDs visible, no rule gives 14, Table 49 lists four risks	ERR
B-39	Med.	p.80	Risk register coverage	No risk for link closure, wheel saturation, dV growth or antenna burial; goal RPNs unquantified	UNS
B-40	Med.	p.11, 19	T-4.6 h deploy / T-4.4 h burn vs Table 2	Impact is 12.5 min before closest approach on the report's timeline; the T-1 h descent is not in Table 2	ERR
B-41	Med.	p.19	Table 2 Umbriel row from an instantaneous 415 m/s burn	No direction gives 4.2 h, 901 m/s and $\geq$ 80 deg together in a 2-body check; the lead model with tides gives 862 m/s, 3.76 h (A-10)	UNS
B-42	Med.	p.15, 43	Triple 8-bit ATmegaS128 for imaging/seismic; 50 GB	No storage/processing element shown; CDH_1/2 TBR	UNS
B-43	Low	p.44	Dose 15-20 krad vs 30 krad part	Margin 1.5-2x; no Uranian-belt dose	ASM

B Reproducibility

The analysis, code and text of this successor study are kept in the PERCI_v2 workspace, and every result is deterministic. The script `analysis/run_all.py` runs every module in dependency order (trajectory sweeps, penetration, thermal, link, power, science, strategies, risk, figures and tables) and the audit, and `pytest analysis/tests` runs 61 tests (analytic limits, energy conservation, linearisation versus nonlinear integration, and determinism). All random draws use fixed seeds (link 20250611, penetration Sobol 2025, thermal Sobol 77, dispersion 11, risk 3). The parameters with provenance tags are in `research/params.json`, the sources in `research/sources.md`, and the contradictions in `research/contradictions.md`.

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Table 3: Novel-Methods Scoreboard. KEEP 4 (N1, N10, N11, N12), PARTIAL 5 (N2, N3, N5, N6, N9), UNPROVEN 1 (N13), FAILED 4 (N4, N7, N8, N14).

Idea	Result (gain over baseline, band)	Cost / caveat	Verdict
N1 Min-ΔV aim at 4.4 h	ΔV −22% (323–328 vs 401–415), $v_{\rm imp}$ −13% (0.75–0.76 vs 0.86), P(acc) 0.94 vs 0.89 in all 4 orientations	arrival 12 min after closest approach; needs 0.6° pointing	KEEP
N2 Earlier release (12 h)	ΔV 121–125 m/s, propulsion 9 kg vs 26–41 kg	P(acc) 0.74–0.85 (line of sight); geometry-dependent	PARTIAL
N3 Release 24–48 h with thruster	ΔV 17–100 m/s, 3–8 kg; descent data volume 5–8× (115–186 vs 22 MB scaling)	P(acc) 0.47–1.00 (24 h), 0.00–0.99 (48 h): flyby-orientation-dependent; hibernation 1–2 days at 0.64 W is cheap (15–31 Wh)	PARTIAL (best prize; Phase-A study)
N4 Burn at periapsis (Oberth)	ΔV 855 m/s at T−0.1 h vs 328 at 4.4 h; max Oberth factor 1.05	—	FAILED
N5 Staged 2+2 SRB ejection	+11.4 m/s (+2.4%) from dropping 6.5 kg of cases	extra separation event; recoil 0.4–1.0 km (B-29)	PARTIAL (not needed at v2 ΔV)
N6 Retarget aiming for low $v_{\rm imp}$	$v_{\rm imp}$ 0.675 vs 0.756 km/s at ΔV = 390 m/s (LOS-feasible); floor 0.54	flagship range 6.9 Mm; −2.5% hard-ice peak g only	PARTIAL (small benefit)
N7 Retro motor before impact (1 STAR 5D)	−140 m/s: 0.78 → 0.64 km/s	+10.2 kg; hard-ice peak g −2.5%, depth −30%; porous g −29%	FAILED for survival (strength-dominated)
N8 Oblique impact / tumbling	—	allowable incidence 3.6° (8 mm wall, 3,000 g)	FAILED
N9 Nose shape (ψ 3–12)	hard ice: peak g 4.4 → 4.1 kg; porous: 1.6 kg → 0.12 kg, depth ×6	trades shock for burial; mass of sharper nose	PARTIAL
N10 Drag flare (report) vs split body	flare: peak g ×4–10	split forebody/aftbody (DS-2, Mars 96 heritage, not new)	KEEP split body
N11 Slow slew instead of 180° flip	wheel 0.06 N m s vs 2.6–4.0 required (43–66×); slow slew needs 0.05 N m s	450 s slew; momentum dump	KEEP
N12 Alternative target moons	see Table 4: Titania reachable (377 m/s, 0.86 km/s); Oberon 1.7 km/s (too fast); Titania-2045 3.0 km/s	Ariel 0.94 km/s; higher impact speeds raise g in porous targets only	KEEP (Umbriel flyby 1 best)
N13 Titania gravity-assist leverage	∼130 m/s equivalent per flyby, 12–14 m/s per 100 km of periapsis shift (65–71 m/s per 1,000 km)	order-of-magnitude bound, not a trajectory; needs the UOP tour ephemeris	UNPROVEN
N14 Two-penetrator seismic network	P(detect the other's impact) 0.20 at 100 km separation (2×119 kg), 0.11 (2×60 kg)	238 kg exceeds the 180 kg limit before propulsion	FAILED

Table 4: Alternative Target Flybys from the UOP Table. Configuration (+,+), release at 4.4 h and 24 h, hard-ice median target. "Prop. mass" uses the SRB-scaled rocket equation. Titania-2045 and Oberon need 75/117 m/s but arrive at 3.0/1.7 km/s.

Flyby (UOP table)	ΔV 4.4 h (m/s)	v_{imp} (km/s)	hard-ice peak g (g)	ΔV 24 h (m/s)	prop. mass 4.4 h SRB-scaled (kg)	flagship elev. +30 min (deg)
Umbriel-1 (2047-06-22)	328	0.76	4200	61	26	19
Umbriel-2 (2047-10-03)	256	0.89	4400	42	20	-1
Ariel (2047-07-29)	257	0.94	4400	50	20	3
Oberon (2047-07-28)	117	1.71	–	21	9	-56
Titania-2047-11-07	377	0.86	4300	82	31	25
Titania-2045-08-11 (report)	75	3.00	–	13	5	-77

Table 5: Penetrator Heritage Compared with PERCI

Mission (year)	Mass [kg]	v_{imp} [m/s]	Design/test [g]	Outcome	Relay / life
Deep Space 2 (x2) (1999) [33, 34]	3.6	160–200	30,000–60,000	lost, no probable cause found	UHF to Mars Global Surveyor; 1-3 d (Li-SOCl2)
Mars 96 penetrators (x2) (1996) [15]	45.0	60–100	500	lost with launch vehicle	orbiter; 1 yr planned
Lunar-A (x2) (2007) [35–37]	13.0	250–300	5,000–8,000	cancelled 2007 (development overrun)	orbiter, contact ∼every 15 d; ∼1 yr
MoonLITE (x4) (2008) [38, 39]	13.0	300	10,000–20,000	Phase A, not flown; Pendine sled trials 2008 survived	orbiter, 30 kbit/day; >= 1 yr
ESA icy-moon penetrator (2013) [13, 40]	20.0	341	24,000	ice sled test, recovered intact	UHF patch under radome (study); 2 wk (study)
PERCI 2025 report (2025) [2]	110.0	900	–	concept	2 GHz to a 1 m dish (not on UOP); 30 min
PERCI v2 (2026)	119.2	750–760	3,000–18,000	concept (this work)	UHF 401 MHz to UOP, 5.4-6.5 Mm; 30 min - 2 h

Mass is the mass at impact. DS-2 includes the aeroshell; MoonLITE is the penetrator without its 23 kg delivery stage; PERCI v2 is CBE (MEV 149 kg). The MoonLITE peak load is quoted as 10,000 or 20,000 g by different sources [38, 39].

Table 6: How Secondary Payloads Have Been Released from a Carrier

Secondary (carrier)	Release mechanism	v_{sep} [m/s]	Coast	Flagship constraint or lesson
Huygens (Cassini) [46, 47]	Spin/Eject Device: 3 springs, 3 pyronuts, helical guides	0.35	20-22 d, timer wake ∼5 h before entry	orbiter deflection 24 m/s 3 d later; 30 kg support equipment stays on orbiter
Galileo probe [48]	explosive nuts + 3 springs; orbiter spun to 10.5 rpm	–	∼147 d, timer wake 6 h before entry	orbiter spin-up before release
Philae (Rosetta) [49–51]	push-off mechanism, backup spring	0.187	∼7 h descent	released although its thruster check had failed
LICIACube (DART) [52]	spring-loaded box on the spacecraft wall	1.1	15 d	released 15 d before impact
Hayabusa2 SCI [53]	separation at 500 m, 40 min timer	–	40 min	mothership retreats ∼1 km and hides behind Ryugu
MINERVA (Hayabusa) [54]	release command	–	-	released at ∼200 m while ascending (planned ∼70 m); escaped
Artemis I CubeSats [55]	spring dispensers in the stage adapter	–	-	1 rpm thermal roll needed or batteries freeze; 5 of 10 partial/full success
UOP atmospheric probe [4]	(study) released after orbit insertion	–	20-30 d, kept warm by LWRHUs	orbiter carries a UHF antenna for the probe relay

Table 7: Post-Impact Data Products, Volumes and Use

Priority	Product	Raw [MB]	Sent [MB]	Used for
1	accel_impact_pulse_100ms	0.2	0.02	Q5.1c strength profile R(z)
2	rtd_8ch	0.432	0.0432	Q8.2f bulk thermal conductivity
3	housekeeping	0.045	0.045	engineering: survival, thermal-model check
4	seismometer_3ax	1.62	0.648	Q5.1c porosity / coda (partial)

Volumes from the store-and-forward model of Section 5.5; descent thumbnails (0.1 MB each, Q12.8a) are sent before impact.

Table 8: Single-String Elements, Heritage Practice and Evidence

Element	PERCI v2	Heritage practice	Model value or datum
Separation	one spring/pyro release (report)	Huygens: 3 pyronuts, 3 springs	heritage TRL 9
Wake/sequencing timer	one CDH	Huygens: triply redundant mission timer	no datum
SRB ignition after 16 yr	4 STAR 5D fired together	no long-soak ignition datum	no datum
Aim (burn + pointing)	spin-stabilised burn, 0.6 deg	-	P(acceptable) = 0.94 (model)
Structure at 0.75 km/s	one body + aftbody	highest ice test 341 m/s	no datum above 341 m/s
Cells + electronics shock	one stack	DS-2: flight lot never impact-tested	no datum
Forebody-aftbody cable	one cable	DS-2 flex cable	no datum
UHF transmitter + antenna	one chain	Huygens: two S-band chains	post-impact >= 0.04 MB in 95% (model)
Flagship relay pointing	UOP UHF LGA at CA+12 min	UOP probe relay (study)	G/T unknown

Table 9: Options for In-Situ Measurements at Umbriel

Option	Wet mass [kg]	Launch equiv. [kg]	In-situ score	Rows met
Flagship remote sensing only	0	0	0.0	none in situ (thermal inertia of the top cm and imagery only)
PERCI v2 (this work)	160–191	420–501	3.0	Q5.1c, Q8.2f achievable; Q12.8a, seismic partial
13 kg-class penetrator	21–31	55–81	2.5	Q5.1c, Q8.2f achievable; seismic partial; no descent imagery beyond a 10 g imager
MASCOT-class soft lander	25–37	66–97	2.0	Q8.2f (contact), surface imagery; no strength profile

Small penetrator: 13 kg body [35, 38] plus 5 kg for aim, avionics and separation (ASSUMED) and a monopropellant stage for 328 m/s (Isp 220 s, 25% tankage, 2.5 kg hardware, as in Table 2), band ±25%. Lander: a 10 kg MASCOT-class body [44] plus 3 kg of terminal-guidance sensors (ASSUMED) and $1.1 \times (328 + 750)$ m/s of braking.

Table 10: Recomputed Risk Register. p is the model probability; the D (detectability) scores are ASSUMED.

ID	Failure mode	S	p report	RPN concept	p v2 design	RPN	report ID
TRJ-01	Impact not acceptable (miss, incidence >10 deg, speed, flagship line of sight)	5	0.11	60	0.06	60	PRP-01
TRJ-02	Delta-V shortfall from mass growth (S-120 band)	5	0.79	75	0.00	15	PRP-01
STR-01	Peak deceleration above 10,000 g (instrument/structure shock) given target uncertainty	5	0.11	75	0.11	60	–
STR-02	Aft antenna buried (nose depth > body length 1.3 m); flare brake raises peak g 4-10x	4	1.00	80	0.15	36	–
COM-01	UHF link does not close (<3 dB margin at 6 kbps, 5.4-6.5 Mm, buried 3-30 m)	4	1.00	40	0.30	32	–
THM-01	Thermal hold for 2 h needs more than 20 W of heater (stresses cells and heater budget)	4	1.00	60	0.12	36	–
PWR-01	Cell stack brownout at cold, high current (4 kg commercial LTC at 140 W, 253 K)	5	1.00	75	0.10	45	–
ADC-01	Reaction wheel saturates in the 10 deg/s post-burn flip	4	1.00	40	0.05	24	–
PRP-02	SRB ignition failure after 13.4 yr cold soak	5	0.05	75	0.05	75	–
MAS-01	Wet mass above 180 kg at MEV (S-120 growth)	4	1.00	20	0.80	20	–
SEI-01	Broadband seismometer does not survive >1,000 g	4	0.50	100	0.40	64	–
SCI-01	Interior-structure science (Q5.1a) not achievable with one 30-min station	3	1.00	15	1.00	15	–

Table 11: Changed Values (Old → New, Reason, Source)

Quantity	Old (report)	New (v2)	Reason	Source
ΔV requirement	450 m/s (476 cap.)	330 m/s; capability 400 at MEV mass	minimum central hit 323–328 m/s over 4 orientations; capability falls with growth (B-23)	Sec. 5.1, A-11
Impact speed	0.90 km/s	0.75–0.76 km/s	min-ΔV solution	Sec. 5.1
Flight time / impact time	4.2 h, before CA	4.6 h, 12 min after CA	min-ΔV geometry	Sec. 5.1
Burn / attitude	retrograde, 10°/s 180° flip	nose-forward burn, 67° slew $\leq$ 0.15°/s	ΔV vector is 67° from impact velocity; wheel 43–66× short (B-28)	A-14, B-28
Impact mass	110 kg	119 CBE (149 MEV)	budget 159−40 (B-27)	B-27
Aft body	single body + drag flare	split forebody/aftbody, surface antenna	flare raises g 4–10× and leaves antenna buried	A-09
Design shock	5,400–10,400 kN, 3.4–8.3 m	distribution; design 10,000 g with isolation for seismometer/cells	Sobol result over 3 classes	Sec. 5.2
Heater	140 W (30 min)	0 W (30 min), $\leq$24W (2 h), preheat 10 W	core 6–28 kJ/K, $G \leq 0.25$ W/K	Sec. 5.3
Cells	192 (4 kg), 4s48p	8–12 LSH20-class, 2 kg	load $\leq$20 W in a warm core; cold pulse current is the limit	Sec. 5.4
Energy	195.9 Wh	$\approx$66 Wh (ASSUMED loads)	imagers 30 min, preheat 1 h	Sec. 5.4
Comms	2 GHz, 5 W DC, 10–20.1 kbps (claim 2.14 Mbps)	UHF 401 MHz, $\leq$6 kbps, deployable antenna; UOP G/T required	no S-band on UOP; link does not close with assumed gains	B-01, Sec. 5.5
Memory / data	50 GB; 20 GB	$\sim$1 GB; $\leq$16 MB returned	link-limited	B-10
Science	5 Decadal Q	Q5.1c, Q8.2f, partial Q12.8a	Sec. 5.6	–
Mass margin	12% (wet vs 180)	MEV 191 kg wet / 163 dry; open	S-120 growth	B-33
Storage before use	13.4 yr (cruise)	16.0 yr (launch to flyby)	cruise plus tour before release	Sec. 6.3
Cruise thermal	0.64 W, no motor heater	1.4–13 W motor heater, or mount inside the UOP enclosure	analogous motor storage floor −6.7°C	Sec. 6.5
Spin-up	on-board SRMs + yo-yo (1.75 kg)	flagship spin/eject device	Huygens heritage	Sec. 6.2
Cost class	SIMPLEx ($55 M cap)	SSCM floor $46–59 M (2024); contributed element	SMAD Table 20-6	Sec. 6.6

Table 12: Technology Readiness Assessment (Ours, Stated Honestly)

Technology (assessment, ours)	TRL	Basis
Penetration at 0.75 km/s into cold ice	2–3	highest test 341 m/s [13]; analytic only
Penetrator structure, 100+ kg, 4,000–18,000 g	3	DS-2 (3 kg, 200 m/s) lost with no cause identified [17]
Broadband seismometer at >1,000 g	3–4	Lunar-A 8,000–10,000 g prototype [20]; MEMS needs protection [21]
Li-$SOCl_2$ cells, cold, impact	4	commercial cells TRL 9 at low rate; DS-2 hardened cell custom [22]
STAR 5D motor	9 / 5	flown; 16 yr storage untested [23]
Guidance/aim from flagship, 0.6°, 20 km	3	analysis only
UHF relay link, buried antenna	3–4	Electra/Proximity-1 relay TRL 9 [28]; buried penetrator antenna not tested
Warm-core insulation under shock	3	aerogel shock survival unverified
Split forebody/aftbody	6 (at 200 m/s)	Mars 96 / DS-2 heritage [15, 16]